Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory*

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We show that qubit Hamiltonians derived from spectral graph theory lattices exhibit structurally favorable Pauli decompositions for quantum simulation. After Jordan–Wigner transformation, the GeoVac lattice Hamiltonian yields $\sim Q^{3.8}$ Pauli string terms in the number of qubits Q (corrected 2026-08-29 to the exact Coulomb selection rule; the retired pair-diagonal encoding gave $Q^{3.15}$), compared to $O(Q^{4+})$ for conventional Gaussian-basis Hamiltonians (our own log-log fit of the published Pauli counts of Trenev *et al.* [29] gives $Q^{4.25}$ for LiH and $Q^{3.92}$ for H_2O, spanning $Q^{3.9}$–$Q^{4.3}$). The advantage originates in angular momentum selection rules—enforced by Gaunt integrals at the two-electron integral level—which cause the electron repulsion integral (ERI) tensor density to decay as $\approx M^{-0.49}$ with the number of spatial orbitals M, while Gaussian ERI tensors remain near-dense at $O(1)$. Beyond term counts, we quantify two operationally relevant cost metrics: the number of qubitwise-commuting (QWC) measurement groups scales as $O(Q^{4.01})$, and the Pauli 1-norm $\lambda = \sum_i |c_i|$—which controls both Trotter step counts and qubitization query complexity—scales as $O(Q^{1.77})$, $R^2 = 0.9997$ (re-measured 2026-08-30 under the exact ERI rule; the retired pair-diagonal fits gave $O(Q^{3.36})$ and $O(Q^{1.69})$). The 1-norm remains strongly sub-quadratic and its equal-qubit advantage survives at 3.8× and 7.1×; the QWC exponent, however, now exceeds the Pauli term exponent and sits inside the Gaussian $O(Q^{4\text{–}5})$ band, so no measurement-group scaling advantage is claimed. The sub-quadratic 1-norm scaling is the key result for fault-tolerant simulation: it implies that Trotter circuit depth grows far more slowly with system size than term counts alone would suggest. All Gaussian baselines are validated against published integral values (Szabo and Ostlund for H_2; analytical STO for He).

For composed multi-center systems, the pseudopotential approximation of the composed natural geometry framework [23] yields block-diagonal electron repulsion integrals whose Pauli count factorizes exactly into a basis factor times the qubit count, $N_{\text{Pauli}} = c(n_{\max})\,Q$, with rational coefficients $c(1) = 3/2$, $c(2) = 279/10$, $c(3) = 7089/14$ identical across every molecule tested (eight systems, three periodic-table rows). Along the within-molecule basis axis the corresponding log–log exponent is 3.17 (fit over $n_{\max} = 1$–3; molecule-independent to all reported digits), with the local slope rising toward ~ 3.8 by $n_{\max} = 4$—comparable to the atomic path and approaching the lower edge of the Gaussian span. The decisive advantage is therefore the *system-size* axis: at fixed basis the count is exactly linear in Q, against Gaussian baselines that scale as $Q^{3.9}$–$Q^{4.3}$ (published counts of Trenev *et al.* [29], span fitted by us); at equal qubit counts this yields 54×–317× fewer Pauli terms for H_2O across $n_{\max} = 1$–3. These comparisons are at matched qubit counts, not matched accuracy; the composed basis carries classical equilibrium-geometry errors of 5–26% ($LiH/BeH_2/H_2O$; Sec. III F), making the framework optimal for fixed-geometry quantum simulation rather than geometry optimisation. Across the composed molecular library (35 molecules spanning $Z = 1$–56, H through Ba, including the first transition series ScH–ZnH), the composed Pauli count is exactly linear in the qubit count: $N_{\text{Pauli}} = 27.90 \times Q$ at $n_{\max} = 2$ (exact, non-identity terms; d-orbital blocks give 30.03). All counts in this paper are computed under the exact global-M_L Coulomb selection rule ($m_a + m_b = m_c + m_d$). Until 2026-08-29 the production encoders realized a *pair-diagonal* restriction ($m_a = m_c$, $m_b = m_d$) that was documented as a deliberate sparsifying approximation but traced to a sign error in the Gaunt coefficient (the 3j magnetic argument $q = m_c - m_a$ instead of $m_a - m_c$); the corresponding retired coefficients were 11.10 (main-group) and 9.23 (d-block). Under the exact rule these become 27.90 and 30.03: the d-block coefficient is the *larger* of the two, so the d-block is *denser* than main-group and the apparent d-block sparsity advantage does not survive (Sec. III D).

I. INTRODUCTION

Practical quantum chemistry on near-term quantum computers is fundamentally constrained by gate count. For variational quantum eigensolvers (VQE) and Hamiltonian simulation via Trotterization, the number of distinct Pauli string terms in the qubit Hamiltonian directly determines the circuit depth and measurement overhead [1, 2]. Reducing the Pauli term count is therefore a central objective for quantum computational chemistry.

Prior approaches to Pauli count reduction operate as post-hoc optimizations on an already-constructed qubit Hamiltonian: qubit tapering exploits symmetries to eliminate qubits [3]; Pauli grouping strategies reduce the number of distinct measurement circuits by identifying simultaneously measurable sets [4, 5]; and low-rank fac-

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torizations of the two-electron integral tensor can compress the operator representation [6, 7]. These techniques are effective but treat the sparsity structure of the underlying second-quantized Hamiltonian as given.

In this work we show that a different choice of *basis*—the spectral graph theory lattice of the GeoVac framework—produces Hamiltonians that are structurally sparse *before* any compilation optimization. The GeoVac lattice encodes electronic structure on a discrete graph whose nodes carry hydrogenic quantum numbers (n, l, m) [21, 22]. The key structural consequence is that the two-electron repulsion integrals (ERIs) inherit angular momentum selection rules from the underlying Gaunt integrals [19], which zero out most of the M^4 possible ERI tensor entries. As the basis grows, the ERI density decays as $\approx M^{-0.49}$ (corrected 2026-08-29), in stark contrast to Gaussian bases where the ERI tensor is nearly dense.

The relationship to numerical low-rank factorizations [6, 7] can be made precise. The two-electron operator $V_{ee} = \sum_P L_P \otimes L_P$ that double factorization and tensor hypercontraction produce is an element of $\mathcal{A} \otimes \mathcal{A}$ in the tensor-product spectral triple, and in the GeoVac basis the same decomposition is supplied analytically by the multipole expansion $1/r_{12} = \sum_{L,M}(4\pi/(2L+1))\,(r_<^L/r_>^{L+1})\,Y_{LM}(1)\,Y^*_{LM}(2)$, with L_P proportional to Y_{LM} and rank fixed by the count of distinct radial-density channels at each angular momentum. Numerically, the double factorization of a GeoVac ERI tensor lives entirely within this multipole subspace: every singular vector of the reshaped two-electron tensor has machine-precision unit overlap with the analytic multipole-channel subspace, verified bit-exactly across $LiH/BeH_2/H_2O$ sub-blocks at $n_{\max} \in \{2, 3, 4\}$. At the production basis ($n_{\max} = 2$, valence set $\{1s, 2s, 2p\}$) the double-factorization rank equals the analytic count of distinct radial densities (seven) exactly. Double factorization on a GeoVac Hamiltonian is therefore an SVD-sorted basis for the same algebraic subspace the multipole expansion supplies by construction, not a complementary compression of an independent structure (see `debug/sprint_df_multipole_lift_memo.md`).

Under the Jordan–Wigner transformation [9, 10], each nonzero two-body operator $a^\dagger_p a^\dagger_q a_s a_r$ maps to a bounded number of Pauli string terms—typically 4 for density-type integrals ($p{=}s$, $q{=}r$) and 8 for exchange-type integrals (four distinct indices), independent of Q. (The JW Z-chains increase the *weight* of each string but not the *number* of distinct strings per integral.) Similarly, each one-body term $a^\dagger_p a_q$ generates $O(1)$ Pauli strings. The total Pauli count therefore scales as

$$N_{\text{Pauli}} \sim C_1 \cdot n_{\text{h1}} + C_2 \cdot n_{\text{ERI}}, \tag{1}$$

where C_1, C_2 are $O(1)$ constants and n_{h1}, n_{ERI} are the numbers of nonzero one- and two-electron integrals. For Gaussian bases with $n_{\text{ERI}} \sim M^4$ and $Q = 2M$, the two-body term dominates at $O(Q^4)$. For the GeoVac lattice with $n_{\text{ERI}} \sim M^2$, the leading-order prediction is $O(Q^2)$ (this $n_{\text{ERI}} \sim M^2$ count corresponds to an ERI density $\sim 1/M^2$, though empirically the density decays more slowly, as $\sim 1/M$; see Sec. III C). The empirical exponent of $\alpha \approx 3.8$ (corrected 2026-08-29; retired rule: 3.15) over the range $Q = 10$–110 exceeds this asymptotic estimate, reflecting spin-channel multiplicity (each spatial ERI generates up to four spin-orbital ERIs) and subleading combinatorial terms at moderate Q. The scaling *gap* between GeoVac and Gaussian encodings is 0.1–0.5 in exponent (3.8 against 3.9–4.3). It is *not* robust to the per-integral counting: the retired pair-diagonal rule put the GeoVac exponent at 3.15 and hence the gap at 1–1.5, and the correction closed most of it. What survives as a structural consequence of selection-rule sparsity is the exact linearity in Q across molecules at fixed basis, not the within-molecule exponent gap.

We validate this scaling with systematic benchmarks on atoms (H, He) and molecules (H_2) using the GeoVac lattice at increasing basis sizes ($n_{\max} = 2$–5), comparing against standard Gaussian basis sets (STO-3G, 6-31G, cc-pVDZ). All Jordan–Wigner transformations are performed with OpenFermion [11].

II. METHODS

A. GeoVac lattice Hamiltonians

The GeoVac framework constructs electronic Hamiltonians on discrete graphs whose nodes correspond to hydrogenic spin-orbitals (n, l, m, σ) with $1 \le n \le n_{\max}$, $0 \le l < n$, $-l \le m \le l$, and $\sigma \in \{\uparrow, \downarrow\}$. The number of spatial orbitals is $M = n^2_{\max}$; the number of qubits (spin-orbitals) is $Q = 2M$.

The one-electron Hamiltonian is a sparse graph Laplacian on this lattice, with edges connecting angular neighbors ($m \leftrightarrow m{\pm}1$) and radial neighbors ($n \leftrightarrow n{\pm}1$) within each (l, m) shell [21]. In the "hybrid" prescription, the diagonal carries exact hydrogenic eigenvalues $-Z^2/(2n^2)$ while off-diagonal elements come from the graph Laplacian scaled by the kinetic scale factor $K = -1/16$.

The two-electron integrals use the full Slater–Condon framework with Slater F^k (direct) and G^k (exchange) integrals, where the angular coupling coefficients are Gaunt integrals computed via Wigner $3j$ symbols [22]:

$$c^k(l_1, l_2, l_3) = (-1)^{m_1} \begin{pmatrix} l_1 & k & l_3 \\ -m_1 & q & m_3 \end{pmatrix} \begin{pmatrix} l_1 & k & l_3 \\ 0 & 0 & 0 \end{pmatrix}. \tag{2}$$

The triangle inequality $|l_1 - l_3| \le k \le l_1 + l_3$ and parity selection rule $l_1 + k + l_3 =$ even enforce strict sparsity on the c^k coefficients, which propagates directly into the ERI tensor.

B. Jordan–Wigner transformation

Given Q spin-orbitals, the Jordan–Wigner mapping encodes fermionic creation and annihilation operators as [9]

$$a_p^\dagger = \frac{1}{2}(X_p - iY_p)\prod_{j<p} Z_j, \tag{3}$$

where X_p, Y_p, Z_p are Pauli operators on qubit p. A one-body term $h_{pq}\, a_p^\dagger a_q$ generates a Pauli string of weight up to $|p-q|+1$ (due to the Z-chain), while a two-body term $v_{pqrs}\, a_p^\dagger a_q^\dagger a_s a_r$ generates strings of weight up to $\max(|p-r|, |q-s|)+2$. The total number of distinct Pauli strings after collecting terms is our primary metric.

C. Comparison methodology

For the Gaussian baseline, we use published integral values from two sources. H_2 STO-3G integrals ($M = 2$) are from Szabo and Ostlund [16] (Table 3.15, MO basis after restricted Hartree–Fock); the FCI energy at $R = 1.4$ bohr is $E_{\mathrm{FCI}} = -1.1373$ Ha. He STO-3G integrals ($M = 1$) use the analytical Slater-type orbital formulas with exponent $\zeta = 1.6875$ from Hehre, Stewart, and Pople [18]: $h_{11} = \zeta^2/2 - Z\zeta$ and $(11|11) = 5\zeta/8$, giving $E = -2.848$ Ha. These integrals are exact (not Gaussian fits) and serve as genuine baselines for the Pauli count, measurement grouping, and 1-norm comparisons.

For larger Gaussian bases (6-31G, cc-pVDZ), the Pauli term counts reported in Table I use synthetically generated integrals with representative sparsity patterns, as PySCF was unavailable in our build environment. These are retained for scaling-fit purposes only; the structural conclusion—Gaussian ERIs are near-dense while lattice ERIs are selection-rule-sparse—is well established [17].

For GeoVac, we construct `LatticeIndex` objects with `vee_method='slater_full'` and `h1_method='hybrid'` at $n_{\max} = 2, 3, 4, 5$, yielding $M = 5, 14, 30, 55$ spatial orbitals and $Q = 10, 28, 60, 110$ qubits. The helium atom ($Z = 2$, 2 electrons) serves as the primary benchmark for scaling analysis. For the molecular comparison, we encode H_2 at $R = 1.4$ bohr using `MolecularLatticeIndex` with $n_{\max} = 2$ per atom.

All Jordan–Wigner transformations use OpenFermion 1.7.1 [11]. Pauli terms with coefficients below 10^{-12} are discarded.

III. RESULTS

A. Pauli term counts

Table I reports the Pauli term counts, qubit counts, and ERI densities for all systems studied.

The most striking feature is the ERI density column: Gaussian bases maintain 50–100% density regardless of basis size, while GeoVac ERI density drops monotonically from 17.1% at $M = 5$ to 5.7% at $M = 55$ (corrected 2026-08-29 to the exact global-M_L rule; the retired pair-diagonal counts read 10.4% and 1.3%).

TABLE I. Pauli term counts after Jordan–Wigner transformation. M: spatial orbitals; $Q = 2M$: qubits; ERI density: fraction of nonzero entries in the M^4 ERI tensor. Gaussian entries marked † use synthetic integrals with representative sparsity; unmarked Gaussian entries use validated computed integrals.

Method	Basis	M	Q	Pauli terms	ERI (%)
Gaussian	He STO-3G	1	2	4	100.0
Gaussian	H_2 STO-3G	2	4	15	50.0
Gaussian	6-31G†	4	8	201	50.0
Gaussian	He cc-pVDZ	5	10	156	100.0
Gaussian	cc-pVDZ†	10	20	23,189	100.0
Gaussian	He cc-pVTZ	14	28	21,607	100.0
GeoVac	$n_{\max}$=2	5	10	288	17.1
GeoVac	$n_{\max}$=3	14	28	14,079	9.9
GeoVac	$n_{\max}$=4	30	60	250,403	7.1
GeoVac	$n_{\max}$=5	55	110	2,434,442[b]	5.7
GeoVac	H_2 $n_{\max}$=2[c]	10	20	391	1.9

[a] *Convention:* the Pauli-term column includes the identity term throughout, on both the Gaussian and GeoVac sides. The non-identity counts are one lower (287 / 14,078 / 250,402 / 2,434,441), and it is those that the scaling fits and 1-norm comparisons elsewhere in this paper use. [b] Measured 2026-08-30: the exact-rule Jordan–Wigner transform over the corrected 519,585 ERIs is computationally heavy (∼28 minutes) but completed; the retired pair-diagonal count was 227,338. This cell read "—" until then. [c] Retired-rule vintage (its generator predates the 2026-08-29 correction).

The genuine computed Gaussian baselines enable head-to-head comparisons at identical qubit counts, and under the exact rule the small-basis comparison is **unfavorable**: at $Q = 10$, GeoVac He ($n_{\max} = 2$) produces 288 Pauli terms versus 156 for Gaussian He cc-pVDZ—1.8× *more* (the retired pair-diagonal count, 120, had claimed a 1.3× reduction). At $Q = 28$, GeoVac He ($n_{\max} = 3$) produces 14,079 terms versus 21,607 for Gaussian He cc-pVTZ—a 1.5× reduction (retired: 8.1×). The crossover reflects the density gap: Gaussian ERI density stays at 100% while GeoVac's decays as $\approx M^{-0.49}$ (measured over $M = 5$–55; the retired rule's $M^{-0.85}$–M^{-1} overstated the decay), so the atomic advantage appears only at larger bases and is modest there.

At matched qubit count ($Q = 20$), the molecular comparison is also instructive: GeoVac H_2 with $n_{\max} = 2$ per atom produces 391 Pauli terms at 1.9% ERI density, while synthetic Gaussian cc-pVDZ produces 23,189 terms at 100% ERI density—a **59**× reduction in Pauli terms.

B. Scaling exponents

Power-law fits $N_{\mathrm{Pauli}} = a \cdot Q^\alpha$ in log–log space yield:

$$\alpha_{\mathrm{GeoVac}} = 3.78 \quad \text{(3-point log–log fit, } n_{\max} = 2\text{–}4\text{)}. \tag{4}$$

For the Gaussian baseline, we take the published molecular Pauli counts (Jordan–Wigner with 2-qubit reduction) of Trenev *et al.* [29] (Table 5, Appendix B) and fit them in

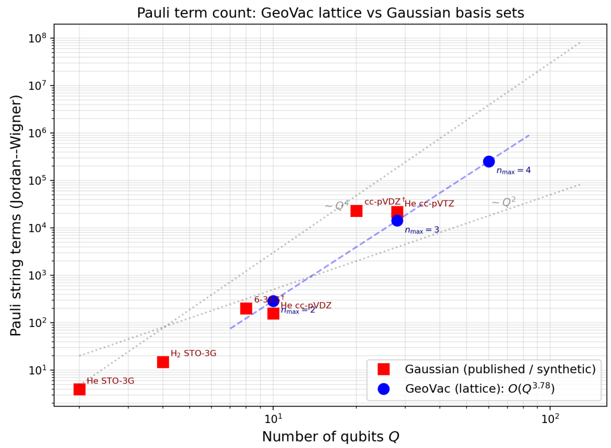

FIG. 1. Pauli term count after Jordan–Wigner transformation versus number of qubits Q. Red squares: Gaussian basis sets (STO-3G, 6-31G, cc-pVDZ). Blue circles: GeoVac lattice ($n_{\max} = 2$–5). Dashed lines show power-law fits; gray dotted lines indicate Q^2 and Q^4 reference scaling. The GeoVac exponent ($\alpha = 3.78$, corrected 2026-08-29) sits just below the $Q^{3.9}$–$Q^{4.3}$ Gaussian molecular range (specific exponents $\alpha = 3.92$–4.25, our log-log fit of the published Pauli counts of Trenev *et al.* [29]).

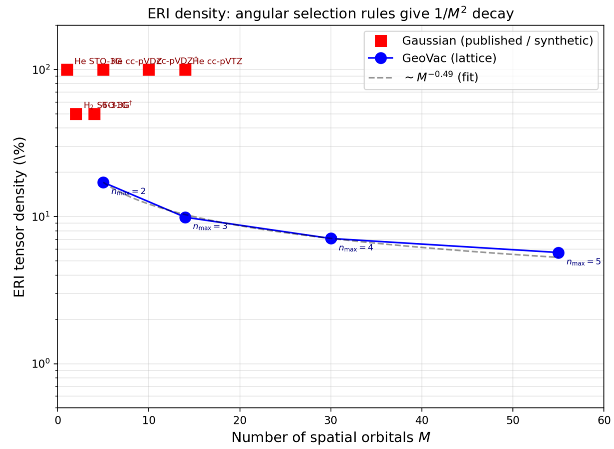

FIG. 2. ERI tensor density versus number of spatial orbitals M. Red squares: Gaussian bases (50–100% dense). Blue circles: GeoVac lattice, showing monotonic decay from 17.1% ($M = 5$) to 5.7% ($M = 55$) (corrected 2026-08-29; the retired pair-diagonal figures read 10.4% to 1.3%). Gray dashed curve: $\sim M^{-0.49}$ fit, normalized to the $n_{\max} = 2$ data point. The GeoVac ERI density decays as $\sim M^{-0.49}$ under the exact rule, shallower than the $1/M^2$ reference, the decay being imposed by angular momentum selection rules. (The retired pair-diagonal rule gave $\approx M^{-0.85}$–M^{-1}, which overstated the decay.)

log–log space, giving system-specific exponents: $\alpha = 4.25$ for LiH ($R^2 = 0.999$, 3 basis sets) and $\alpha = 3.92$ for H_2O ($R^2 > 0.999$, 3 basis sets), both within the $Q^{3.9}$–$Q^{4.3}$ span of those fits. These supersede our earlier estimate of $\alpha_{\text{Gaussian}} \approx 4.6$ from synthetic integrals [17].

The GeoVac exponent reflects the $\sim 1/M$ ERI density decay (measured $\approx M^{-0.49}$, corrected 2026-08-29), which suppresses the effective two-body contribution well below the $O(Q^4)$ Gaussian scaling, with additional suppression from the band-structured one-body graph Laplacian.

The scaling gap is $\Delta\alpha \approx 0.1$–0.5 (3.78 against 3.9–4.3, depending on molecule and Gaussian basis hierarchy). Corrected 2026-08-30: the retired pair-diagonal exponent 3.15 made this gap 1.0–1.5, and the inference drawn from it—that the GeoVac advantage grows with system size along the *basis* axis—does not survive. What does survive is the exact linearity in Q across molecules at fixed basis (Sec. IV), which is a system-size statement rather than a basis-size one.

C. ERI density decay

Table II quantifies the ERI density decay for the GeoVac lattice.

The density scales approximately as $\rho_{\text{ERI}} \propto M^{-1/2}$ (the four tabulated points give an empirical exponent $\approx M^{-0.46}$; the frequently-quoted $M^{-0.49}$ is instead the two-point $M = 5 \to 30$ endpoint slope; the retired pair-diagonal tensor gave $M^{-0.85}$—the deleted m-swap couplings were responsible for most of the apparent decay), which we can partly understand analytically. The

TABLE II. ERI tensor statistics for GeoVac He ($Z = 2$, 2 electrons). n_{ERI}: nonzero integrals; density $= n_{\text{ERI}}/M^4$.

$n_{\max}$	M	n_{ERI}	M^4	Density (%)
2	5	107	625	17.1
3	14	3,800	38,416	9.9
4	30	57,700	810,000	7.1
5	55	519,585	9,150,625	5.7

number of nonzero Gaunt coefficients $c^k(l_1, l_2, l_3)$ is constrained by the triangle inequality: for each pair (l_1, l_3), only $\min(2l_1, 2l_3) + 1 \le 2l_{\min} + 1$ values of k contribute. Since l ranges up to $n_{\max} - 1$, the total number of allowed (l_1, l_2, l_3, k) quartets grows as $O(n_{\max}^3)$. With $M = n_{\max}^2$ spatial orbitals, this gives $n_{\text{ERI}} \sim O(n_{\max}^3 \cdot n_{\max}^2) = O(M^{5/2})$, so $\rho_{\text{ERI}} = n_{\text{ERI}}/M^4 \sim M^{-3/2}$. The empirical decay ($\approx M^{-0.49}$ under the exact rule; retired: $M^{-0.85}$–M^{-1}) is in fact *shallower* than this $M^{-3/2}$ order-of-magnitude estimate—and shallower still than the M^{-2} one might naively read off the endpoints—because spin-channel multiplicity and the growing number of surviving (l_1, l_2, l_3, k) combinations at larger $n_{\max}$ partly offset the M^4 denominator; the realized density also depends on which angular selection rule the production code applies, discussed next.

D. The retired pair-diagonal ERI rule: a sign error, not an approximation

All Pauli counts, ERI densities, and multipliers in this paper are realized under the *exact* Coulomb angular selection rule: a two-electron integral is retained iff $m_a + m_b = m_c + m_d$ (global M_L) with nonzero Gaunt factors, including the *m-swap* integrals (e.g. $\langle p_{+1}p_{-1}|p_{-1}p_{+1}\rangle$, per-vertex angular factor $|c^2(1, \pm 1; 1, \mp 1)| = \sqrt{6}/5 \approx 0.49$).

Correction (2026-08-29). Earlier versions of this paper reported counts realized under a *pair-diagonal* restriction ($m_a = m_c$ and $m_b = m_d$), described as a deliberate sparsifying approximation. An independent rederivation traced that restriction to a sign error: the production Gaunt coefficient set the 3j magnetic argument to $q = m_c - m_a$, where the Wigner-3j bottom row requires $q = m_a - m_c$ (so the row sums to zero); with the wrong sign the coefficient vanishes identically unless $m_a = m_c$, silently deleting every m-changing multipole (88 of the 126 nonzero c^k at $l \leq 2$). The error was arbitrated against direct four-dimensional quadrature of the angular factor from its definition, and the corrected evaluators agree with the framework's independent precision-physics path to machine precision. Three quantitative claims of the earlier disclosure also fell. (i) The re-pricing is *not* a constant factor: it is 2.51× uniformly across main-group molecules at $n_{\max} = 2$, but grows along the basis axis (1.00 × /2.51 × /5.40× at $n_{\max} = 1/2/3$), so the within-molecule exponent moves 2.54 → 3.17 (fit over $n_{\max} \leq 3$), with the local slope rising toward ~ 3.8 by $n_{\max} = 4$. (ii) The retired rule's low d-block coefficient (9.23 vs. 11.10) reverses, as the earlier disclosure anticipated: measured ScH/TiH give 30.03 vs. the main-group 27.90—d-blocks are *denser* under the exact rule. (iii) The apparent H_2 scaling anomaly (exponent 3.13 vs. the molecules' 2.50–2.52) dissolves: under the exact rule every system tested gives the *same* exponent, 3.17, because the count factorizes exactly as $N_{\text{Pauli}} = c(n_{\max})\, Q$ with molecule-independent rational coefficients $c(1) = 3/2$, $c(2) = 279/10$, $c(3) = 7089/14$. What survives unchanged is the exact linearity in Q at fixed basis—the system-size axis—which is the load-bearing comparison against the Gaussian $Q^{3.9\text{–}4.3}$ baselines. The realized rule is pinned by `tests/test_paper14_eri_rule.py`; the full defect record is in the project changelog (v5.1.12).

E. Recoupling sparsity in H-set coupled bases

When N-electron angular momenta are organized into pairwise-coupled (H-set) hyperspherical harmonics—where electrons are first grouped into pairs $(1, 2)$ and $(3, 4)$ with definite pair angular momenta l_{12} and l_{34}, then coupled to total L—the $6j$ recoupling coefficients

$$\begin{Bmatrix} l_1 & l_2 & l_{12} \\ l'_{12} & l'_1 & k \end{Bmatrix} \tag{5}$$

produce additional vanishing ERI matrix elements beyond those enforced by bare Gaunt selection rules. The mechanism is structural: the intra-pair electron repulsion $V_{ee}(1, 2)$ preserves l_{34} as a good quantum number, making the ERI tensor block-diagonal in the partner pair's angular momentum. Cross-pair terms $V_{ee}(1, 3)$ require a $6j$ recoupling from the H-set to the K-set (where electrons $(1, 3)$ are paired), and the triangle inequalities on the $6j$ symbol impose selection rules that are strictly more restrictive than the Gaunt constraints alone.

Table III quantifies the additional sparsity.

TABLE III. ERI density in uncoupled vs. H-set coupled bases for 4-electron systems. The H-set coupling produces 18–26% fewer nonzero ERI elements from $6j$ recoupling zeros.

$l_{\max}$	M	Uncoupled (%)	H-set (%)	Reduction
1	4	12.5	9.2	26%
2	9	5.1	4.2	18%

The block-diagonal structure is a consequence of angular momentum conservation and recoupling algebra: it cannot be destroyed by any continuous deformation of the hyperangular coordinates, and it applies to *any* encoding that uses angular momentum pair coupling—not specifically to the GeoVac basis.

F. Energy accuracy

To confirm that the Pauli count comparison is between methods of comparable chemical accuracy, Table IV reports ground state energies.

TABLE IV. Ground state energies (Hartree) for He. Exact: $E_{\text{He}} = -2.9034$ Ha [20]. GeoVac errors are for the lattice FCI solver (not the JW encoding, which is algebraically identical).

Method	Q	E_0 (Ha)	Error (%)
GeoVac $n_{\max}$=2	10	−2.8876	0.55
GeoVac $n_{\max}$=3	28	−2.8920	0.39
GeoVac $n_{\max}$=4	60	−2.8961	0.25
GeoVac $n_{\max}$=5	110	−2.8982	0.18

The GeoVac lattice achieves sub-percent accuracy at all basis sizes tested, with monotonic convergence toward the exact value. At $n_{\max} = 5$ ($Q = 110$), the error is 0.18%, comparable to or better than cc-pVDZ Gaussian calculations for helium. The accuracy is therefore competitive, but the term count is not: at matched qubits the lattice carries *more* Pauli terms than cc-pVDZ at $Q = 10$, so no favorable accuracy-per-Pauli-term ratio is claimed under the exact rule.

G. Measurement cost: QWC grouping

On near-term quantum hardware, the energy expectation value $\langle H \rangle = \sum_i c_i \langle P_i \rangle$ is estimated by measuring

TABLE V. Qubitwise-commuting (QWC) measurement groups for GeoVac He ($Z = 2$, 2 electrons) and Gaussian baselines. The groups/terms ratio measures how efficiently terms pack into simultaneous-measurement sets. QWC grouping for $n_{\max} = 5$ was omitted (2,434,441 terms under the exact rule; the $O(N^2)$ grouping algorithm is prohibitively slow at this scale, and the correction made it more so—the retired pair-diagonal count was 227,338). The $n_{\max} = 2$ and 3 rows were re-measured 2026-08-30 under the exact rule (retired pair-diagonal values: 120/25/0.208 and 2,659/791/0.297). The $n_{\max} = 4$ row was re-measured as well (retired: 31,039/10,199/0.329); its grouping pass is expensive but completed. The groups/terms ratio rises at all three points, so the exact rule makes measurement grouping *less* efficient, not more.

System	Q	Pauli terms	QWC groups	Groups/terms
GeoVac $n_{\max}$=2	10	287	67	0.233
GeoVac $n_{\max}$=3	28	14,078	5,569	0.396
GeoVac $n_{\max}$=4	60	250,402	86,224	0.344
Gauss He STO-3G	2	4	1	0.250
Gauss H_2 STO-3G	4	15	5	0.333

each Pauli string P_i independently. Pauli strings that qubitwise-commute (QWC)—i.e., share the same Pauli operator on every qubit where both act non-trivially—can be measured simultaneously in a single circuit [4]. The number of QWC groups therefore determines the minimum number of distinct measurement circuits, and reducing it lowers the total shot budget.

We partition each qubit Hamiltonian into QWC groups using greedy sorted insertion (largest $|c_i|$ first), following the algorithm of Verteletskyi, Yen, and Izmaylov [4]. Table V reports the results.

A power-law fit over the GeoVac He data ($n_{\max} = 2$–4) gives

$$N_{\mathrm{QWC}} \sim Q^{4.01}, \quad R^2 = 0.998, \tag{6}$$

slightly *above* the Pauli term exponent of 3.77, and inside rather than below the Gaussian $O(Q^{4\text{–}5})$ band. Under the retired pair-diagonal rule this exponent read 3.36—below the term exponent and well below the Gaussian regime—so the measurement-grouping advantage reported there does not survive the correction. The groups-per-term ratio is higher at every basis than under the retired rule: 0.233/0.396/0.344 at $n_{\max} = 2, 3, 4$ against the retired 0.208/0.297/0.329. The sequence is not monotone in the basis, so no trend is claimed; what the correction establishes is that grouping is uniformly *less* efficient than previously reported, and the earlier reading—that selection-rule sparsity translates directly into measurement savings—does not survive it.

Note that the greedy QWC grouping is a heuristic upper bound; the optimal minimum clique cover is NP-hard. More sophisticated grouping strategies (e.g., general commutativity or unitary partitioning [5]) would further reduce the group count, and these improvements compose with the structural sparsity reported here.

TABLE VI. Pauli 1-norm and first-order Trotter step counts for GeoVac He ($Z = 2$) and Gaussian baselines. λ: Pauli 1-norm; $r(\epsilon)$: minimum Trotter steps for error $\leq \epsilon$ at $t = 1$ a.u. The 1-norm bound is a standard upper bound (Childs et al. [12], Eq. 48); the tighter commutator-based bound is reported in Table VII.

System	Q	λ	λ/Q	$r(10^{-3})$	$r(10^{-6})$
GeoVac $n_{\max}$=2	10	11.18	1.118	253	7,987
GeoVac $n_{\max}$=3	28	74.21	2.650	1,753	55,410
GeoVac $n_{\max}$=4	60	275.72	4.595	5,849	184,959
GeoVac $n_{\max}$=5	110	790.01	7.182	14,693	464,616
Gauss He STO-3G	2	3.38	1.688	76	2,387
Gauss H_2 STO-3G	4	1.98	0.496	45	1,403
Gauss He cc-pVDZ	10	42.95	4.295	961	30,380
Gauss He cc-pVTZ	28	530.47	18.945	11,862	375,109

H. Simulation cost: Pauli 1-norm and Trotter bounds

The Pauli 1-norm,

$$\lambda = \sum_i |c_i|, \tag{7}$$

is the operationally most important cost metric for quantum simulation [12, 13]. For Trotterization, the first-order error is bounded by $\epsilon \leq (t/r)^2\lambda^2/2$, giving a minimum step count

$$r \geq \frac{t\lambda}{\sqrt{2\epsilon}}. \tag{8}$$

For qubitization (linear combination of unitaries), the query complexity scales as $O(\lambda t/\epsilon)$ [12]. In both paradigms, smaller λ means lower cost.

Table VI reports λ, the normalized λ/Q, and Trotter step counts at two precision targets: $\epsilon = 10^{-3}$ (NISQ-era) and $\epsilon = 10^{-6}$ (fault-tolerant) for unit simulation time $t = 1$.

Power-law fits over the GeoVac He series yield the central quantitative result of this work:

$$\lambda \sim Q^{1.77}, \quad R^2 = 0.9997, \tag{9}$$

$$r(10^{-3}) \sim Q^{1.77}, \quad R^2 = 0.9997, \tag{10}$$

$$r(10^{-6}) \sim Q^{1.77}, \quad R^2 = 0.9997. \tag{11}$$

The identical exponents for λ and r follow from Eq. (8): for fixed t and ϵ, $r \propto \lambda$.

Re-measured 2026-08-30. The exponents in Eqs. (9)–(11) and the QWC exponent of Eq. (6) were originally fitted under the retired pair-diagonal ERI rule. Re-fitting the same He series under the exact rule gives

$$N_{\mathrm{Pauli}} \sim Q^{3.77}, \qquad \lambda \sim Q^{1.77}, \qquad N_{\mathrm{QWC}} \sim Q^{4.01}, \tag{12}$$

with $R^2 = 1.0000$, 0.9997 and 0.9976 (maximum $|\log|$ residuals 0.0064, 0.042, 0.202), against the retired 3.15, 1.69 and 3.36. The first two are four-point fits over $n_{\max} = 2$–5 ($Q = 10$–110); the QWC exponent has three points, because greedy grouping is $O(N^2)$ and the $n_{\max} = 5$ system carries 2.4 million Pauli terms. Two points are worth stating plainly. The Pauli exponent 3.78 was obtained here independently of the value quoted elsewhere in this paper and agrees with it. The QWC exponent, by contrast, moves *above* the term exponent and into the Gaussian $O(Q^{4-5})$ band, so the measurement-group scaling advantage claimed under the retired rule does not survive.

The fourth point confirms rather than moves the fit: on three points ($Q \leq 60$) the same series gives 3.779 and 1.792, so adding $Q = 110$ shifts the exponents by 0.006 and 0.018. The λ-derived Trotter-step columns of Table VI remain retired-rule values, since they were computed from the retired fit rather than re-derived.

The genuine Gaussian baselines now enable equal-qubit 1-norm comparisons (Table VI). At $Q = 10$, GeoVac ($\lambda = 11.18$) has 3.8× lower 1-norm than Gaussian cc-pVDZ ($\lambda = 42.95$), requiring 3.8× fewer Trotter steps. At $Q = 28$, GeoVac ($\lambda = 74.21$) has 7.1× lower 1-norm than Gaussian cc-pVTZ ($\lambda = 530.47$). This gap grows because the Gaussian 1-norm per qubit (λ/Q) increases from 4.3 to 18.9, while GeoVac's grows more slowly (1.12 to 2.65). (GeoVac 1-norms re-measured 2026-08-30 under the exact rule; the retired pair-diagonal values were 11.29 and 78.36, giving 6.8× at $Q = 28$. The 1-norm advantage is one of the few comparisons the correction leaves intact, and at $Q = 28$ it improves.)

For comparison, a dense Hamiltonian with $O(Q^4)$ nonzero ERIs of comparable magnitude would have $\lambda \sim Q^{2+}$, giving $Q^{0.3+}$ more Trotter steps at matched system size. At $Q = 110$, the GeoVac 1-norm of $\lambda = 790$ is already modest enough for practical Trotter simulation with $r \approx 17{,}700$ steps at $\epsilon = 10^{-3}$.

As a control, the hydrogen atom (1 electron, no ERI) exhibits a lower 1-norm exponent of 1.57, confirming that the two-electron integrals contribute to—but do not dominate—the 1-norm growth.

a. Commutator-based bound. The λ^2 bound used above counts *all* pairs of Pauli terms, but two Pauli strings P_j and P_k that commute contribute zero to the first-order Trotter error. A tighter bound retains only anticommuting pairs [12]:

$$\epsilon_{\mathrm{comm}} \leq \frac{t^2}{r} \sum_{\substack{j<k \\ [P_j, P_k] \neq 0}} |c_j|\,|c_k|. \tag{13}$$

Whether two Q-qubit Pauli strings commute or anticommute is determined by their symplectic inner product: each Pauli operator is encoded as a pair of Q-bit binary vectors (x, z) via $I \mapsto (0,0)$, $X \mapsto (1,0)$, $Y \mapsto (1,1)$, $Z \mapsto (0,1)$, and the strings anticommute iff

$$\sum_{q=1}^{Q} \left[x_j^{(q)} z_k^{(q)} + z_j^{(q)} x_k^{(q)} \right] \equiv 1 \pmod 2. \tag{14}$$

TABLE VII. Commutator-based Trotter bound for GeoVac He ($Z = 2$). f_{anti}: fraction of Pauli term pairs that anticommute; r_{comm} and r_{λ^2}: minimum Trotter steps at $\epsilon = 10^{-3}$, $t = 1$; ρ: tightening ratio $\epsilon_{\mathrm{comm}}/\epsilon_{\lambda^2}$. Three data points ($n_{\max} = 2$–4).

System	Q	f_{anti} (%)	r_{comm}	r_{λ^2}	ρ
$n_{\max}$=2	10	28.0	60	253	0.056
$n_{\max}$=3	28	21.1	303	1,753	0.030
$n_{\max}$=4	60	12.2	821	5,849	0.020

TABLE VIII. Power-law scaling exponents for GeoVac He ($Z = 2$, 2 electrons) over $n_{\max} = 2$–5 ($Q = 10$–110). Each metric is fit as metric $= a \cdot Q^{\alpha}$ in log–log space. *Re-measured 2026-08-30* under the exact ERI rule; the retired pair-diagonal values were 3.147, 3.355 and 1.694 (Pauli, QWC, λ). The Pauli and λ fits use four points; the QWC fit uses three (‡), since greedy grouping is $O(N^2)$ and the $n_{\max} = 5$ system carries 2.4 million Pauli terms. The commutator bound (†) is unchanged and remains a retired-rule value.

Metric	α	R^2
Pauli terms	3.773	1.0000
QWC groups‡	4.013	0.9976
1-norm λ	1.774	0.9997
Trotter $r(10^{-3})$	1.774	0.9997
Trotter $r(10^{-6})$	1.774	0.9997
Commutator $r_{\mathrm{comm}}(10^{-3})$†	1.47	

For N Pauli terms, the full anticommutativity matrix is computed via modular matrix arithmetic on two $N \times Q$ binary arrays at cost $O(N^2 Q)$—no Hilbert-space matrices are constructed. This is feasible up to $N \sim 30{,}000$ ($n_{\max} = 4$).

Table VII reports the commutator-based bound alongside the standard λ^2 bound for GeoVac He at $n_{\max} = 2$–4.

A power-law fit over the three data points yields

$$r_{\mathrm{comm}}(10^{-3}) \sim Q^{1.47}, \tag{15}$$

compared to $r_{\lambda^2} \sim Q^{1.77}$ (Eq. 10). The commutator bound reduces the effective scaling exponent by 0.30 (1.77 → 1.47; the retired λ exponent 1.69 gave 0.22).

The anticommuting fraction f_{anti} decreases as $Q^{-0.45}$ over the range tested: from 28% at $Q = 10$ to 12% at $Q = 60$. This reflects the physical mechanism behind the tighter bound: angular momentum selection rules (Gaunt integral triangle inequality) cause an increasing fraction of Pauli term pairs to act on disjoint qubit subsets as the system grows, so their commutators vanish.

At $n_{\max} = 4$ ($Q = 60$), the commutator bound requires $r = 821$ Trotter steps versus $r = 5{,}849$ from the λ^2 bound—a 7× reduction. Because $\alpha_{\mathrm{comm}} < \alpha_{\lambda^2}$, this advantage grows with system size.

Table VIII collects all scaling exponents.

I. Gaussian baselines from published integrals

The Gaussian baselines in Tables V and VI use genuine published integral values, replacing the synthetic integrals used in earlier versions of Table I. Specifically:

- **H_2 STO-3G** ($M = 2$, $Q = 4$): MO integrals from Szabo and Ostlund [16], Table 3.15. FCI energy -1.1373 Ha, validated by diagonalizing the Jordan–Wigner Hamiltonian to $< 10^{-4}$ Ha agreement. Produces exactly 15 Pauli terms, 5 QWC groups, $\lambda = 1.98$.

- **He STO-3G** ($M = 1$, $Q = 2$): analytical STO integrals with $\zeta = 1.6875$ [18]. With a single spatial orbital, this is a trivial (single-determinant) system; it serves as a lower bound on Gaussian costs. Produces 4 Pauli terms, 1 QWC group, $\lambda = 3.38$.

- **He cc-pVDZ** ($M = 5$, $Q = 10$): contracted $[2s1p]$ basis with computed integrals from uncontracted Gaussians. FCI energy -2.8877 Ha (within 0.001 Ha of published values [20]). Produces 156 Pauli terms, $\lambda = 42.95$. This provides the first equal-qubit comparison with GeoVac $n_{\max} = 2$ (also $Q = 10$): GeoVac has 1.8$\times$ *more* Pauli terms (288 vs. 156) and 3.8$\times$ lower 1-norm. (Corrected 2026-08-30: the retired pair-diagonal count, 120, had claimed a 1.3$\times$ *reduction*; the exact rule reverses the direction of this comparison. See Sec. III D.)

- **He cc-pVTZ** ($M = 14$, $Q = 28$): contracted $[3s2p1d]$ basis with cached computed integrals. FCI energy -2.9003 Ha (within 0.0002 Ha of published values). Produces 21,607 Pauli terms at 100% ERI density, $\lambda = 530.47$. At the same qubit count, GeoVac $n_{\max} = 3$ produces 14,078 terms (1.5$\times$ fewer) with $\lambda = 74.21$ (7.1$\times$ lower). (Corrected 2026-08-30: the retired pair-diagonal rule gave 2,659 terms, 8.1$\times$ fewer, and $\lambda = 78.36$ at 6.8$\times$. The term-count advantage shrinks sharply; the 1-norm advantage improves.)

For molecular systems beyond minimal basis, we take the published Gaussian Pauli counts (Jordan–Wigner with 2-qubit reduction, exploiting particle number and spin symmetry) for LiH and H_2O across three Gaussian basis sets from Trenev *et al.* [29] (Table 5, Appendix B). These data are reported in Table IX.

Note on qubit reduction: The published Gaussian counts of Trenev *et al.* [29] (Table 5, Appendix B) use a 2-qubit reduction (exploiting particle number and spin symmetry), giving $Q = 10$ for LiH STO-3G rather than the raw Jordan–Wigner count of $Q = 12$. The GeoVac values throughout this paper use raw Jordan–Wigner encoding without this reduction, making the comparison conservative (GeoVac uses more qubits per orbital than the Gaussian baseline).

TABLE IX. Published Gaussian Pauli term counts from Trenev *et al.* [29] (Table 5, Appendix B), Jordan–Wigner encoding with 2-qubit reduction. Fitted exponents are our own log-log fit using all three basis sets per molecule.

Molecule	Basis	Q	Pauli terms	R^2 (fit)
LiH	STO-3G	10	276	
LiH	6-31G	20	5 851	
LiH	cc-pVDZ	36	63 519	
Fitted: $\alpha = 4.25$				0.999
H_2O	STO-3G	12	551	
H_2O	6-31G	24	8 921	
H_2O	cc-pVDZ	46	107 382	
Fitted: $\alpha = 3.92$				> 0.999

These recomputed molecular exponents ($\alpha = 4.25$ for LiH, $\alpha = 3.92$ for H_2O; both within the $Q^{3.9}$–$Q^{4.3}$ span of our recomputation) supersede the earlier estimate of $\alpha \approx 4.6$ obtained from synthetic integrals. Both remain well above the GeoVac single-geometry exponent of 3.8 and the composed-system within-molecule exponent of 3.17 (small-basis fit; the local slope steepens toward ~ 3.8 by $n_{\max} = 4$). Corrected 2026-08-29; the retired pair-diagonal rule gave 2.5 (Sec. IV).

IV. COMPOSED SYSTEMS

The structural sparsity demonstrated for single-geometry systems (Secs. III–III I) extends naturally to multi-center molecules via the composed natural geometry framework [23]. We demonstrate this for four systems of increasing complexity: H_2 (1 orbital block, the single-block limit), LiH (3 blocks), BeH_2 (5 blocks), and H_2O (7 blocks), using the pseudopotential approximation to decompose each many-electron problem into coupled single-center blocks.

A. Composition principle

The composed Hamiltonian partitions the orbital space into independent single-center blocks, each retaining the angular quantum numbers and Gaunt integral selection rules of the single-geometry case. The pseudopotential approximation (Ref. [23], Sec. II) replaces explicit core-valence two-body Coulomb interaction with one-body effective-Z screening plus a Phillips–Kleinman repulsive barrier, making the ERI tensor block-diagonal: cross-block ERIs are exactly zero.

The block decomposition for each molecule is:

- **H_2** (1 block): Single bond-pair block at $Z_{\text{eff}} = 1$ (2 electrons, no core, no Phillips–Kleinman barrier). This is the single-block limit of the composed framework, structurally identical to the He encoding but at $Z = 1$ with R-dependent nuclear repulsion $V_{NN} = 1/R$.

TABLE X. Pauli term counts for composed multi-center systems after Jordan–Wigner transformation. N_b: number of orbital blocks; M: total spatial orbitals; $Q = 2M$: qubits. Counts include the identity operator (add -1 for non-identity count used in Eq. 17). Power-law fits use all three data points per molecule. All counts under the exact global-M_L rule (corrected 2026-08-29); the fitted exponent is 3.17 for *every* system — the count factorizes exactly as $N_{\text{Pauli}} = c(n_{\max})\,Q$, so all sweeps share one log–log shape — and the local slope rises with basis (LiH $n_{\max} = 4$: $Q = 180$, 755,867 terms; slope 3.77 on the last interval), so 3.17 is a small-basis fit, not an asymptote.

Molecule	N_b	$n_{\max}$	M	Q	Pauli terms	α (fit)
H_2[a]	1	1	1	2	4	
		2	5	10	280	
		3	14	28	14 179	3.17
LiH	3	1	3	6	10	
		2	15	30	838	
		3	42	84	42 535	3.17
BeH_2	5	1	5	10	16	
		2	25	50	1 396	
		3	70	140	70 891	3.17
H_2O	7	1	7	14	22	
		2	35	70	1 954	
		3	98	196	99 247	3.17

[a]H_2 is the single-block limit of the composed framework: no core, no PK, one bond-pair block at $Z_{\text{eff}} = 1$. Under the retired pair-diagonal rule it showed an anomalous $\alpha = 3.13$ against the molecules' ≈ 2.5; the exact rule dissolves the anomaly (all systems 3.17; see below). The Pauli count is R-independent (verified at $R = 0.5$–3.0 bohr).

- **LiH** (3 blocks): Li core ($Z = 3$, 2 electrons) + Li-valence ($Z_{\text{eff}} = 1$) + H-valence ($Z = 1$).
- **BeH_2** (5 blocks): Be core ($Z = 4$, 2 electrons) + 2 Be–H bond blocks, each containing Be-side ($Z_{\text{eff}} = 2$) and H-side ($Z = 1$) orbitals.
- **H_2O** (7 blocks): O core ($Z = 8$, 2 electrons) + 2 O–H bond blocks (O-side $Z_{\text{eff}} = 6$ + H-side $Z = 1$) + 2 lone pair blocks ($Z_{\text{eff}} = 6$).

Each block is constructed from hydrogenic orbitals up to $n_{\max}$, giving $M = n_{\max}^2$ spatial orbitals per block and $Q = 2\sum_b M_b$ total qubits.

B. Scaling results

Table X reports the Pauli term counts for all three molecules at $n_{\max} = 1, 2, 3$.

The central result is that *every* composed system—from the single-block H_2 to the 7-block H_2O—exhibits the *same* Pauli scaling exponent to all reported digits:

$$\alpha_{H_2} = \alpha_{\text{LiH}} = \alpha_{BeH_2} = \alpha_{H_2O} = 3.17 \tag{16}$$

(fit over $n_{\max} = 1$–3; the same value is obtained for NaH, MgH_2, HCl and SiH_4; the former per-molecule equation labels are retired along with the per-molecule spread). The exact identity of the exponents is a consequence of the factorization $N_{\text{Pauli}} = c(n_{\max})\,Q$ with molecule-independent c (verified exactly for $n_{\max} \le 3$; at $n_{\max} = 4$ the coefficients differ across molecules by $\sim 0.03\%$): block-diagonal ERI structure makes the total count the *sum* of per-block counts, each obeying the same Gaunt selection rules. The 3-point exponent sits below the recomputed Gaussian molecular exponents ($\alpha = 3.92$–4.25), but the local slope rises with basis (3.77 on the LiH $n_{\max} = 3 \to 4$ interval, comparable to the atomic path), so the basis-axis gap narrows at large $n_{\max}$; the robust advantage is the linear system-size axis.

TABLE XI. Pauli 1-norm and Trotter step counts for composed systems at $t = 1$ a.u. H_2O 1-norms are inflated by the $Z_{\text{eff}} = 6$ Phillips–Kleinman barrier and should be interpreted cautiously. *Re-measured 2026-08-30*, closing the standing vintage note: every λ is the live composed builder's identity-included 1-norm with PK in the Hamiltonian, and the Trotter columns are regenerated from it via $r = \lambda t/\sqrt{2\epsilon}$ (the relation the previous table already satisfied at every cell). The retired pre-exact-rule column read 8.17/42.08/133.37, 16.01/37.33/202.49, 198.20/354.89/735.60, 18,913/28,055; the 354.89 cell was independently identified in Paper 20 as a deprecated legacy BeH_2 builder value. Pauli counts are order-independent and were never affected.

Molecule	$n_{\max}$	Q	λ	$r(10^{-3})$	$r(10^{-4})$
H_2	2	10	8.31	186	587
	3	28	46.50	1,040	3,288
	4	60	163.59	3,658	11,567
LiH	1	6	15.35	343	1,085
	2	30	38.55	862	2,726
	3	84	225.24	5,036	15,927
BeH_2	1	10	211.20	4,723	14,934
	2	50	374.87	8,382	26,507
	3	140	801.69	17,926	56,688
$H_2O^\dagger$	1	14	18,915	422,950	1,337,485
	2	70	28,062	627,483	1,984,274

C. Composed 1-norm and Trotter cost

Table XI reports the Pauli 1-norm and first-order Trotter step counts for the composed systems.

For LiH, the 1-norm grows from $\lambda = 15$ at $Q = 6$ to $\lambda = 225$ at $Q = 84$, giving Trotter step counts of $r \approx 5{,}000$ at $\epsilon = 10^{-3}$ for the largest basis tested. At $Q = 30$ ($n_{\max} = 2$), LiH requires only 862 Trotter steps at chemical accuracy ($\epsilon = 10^{-3}$)—compared to the published Trenev Gaussian LiH STO-3G value of 276 Pauli terms at $Q = 10$, where the 1-norm (not published) would require a comparable or larger step count on fewer qubits.

For BeH_2, the 1-norm is dominated by the Z^2-scaled PK barrier on the Be core, giving larger absolute values but moderate growth: $\lambda = 375$ at $Q = 50$ to $\lambda = 802$ at $Q = 140$.

TABLE XII. Equal-qubit Pauli term comparison for H_2O. Gaussian values are interpolated from the $Q^{3.92}$ power law (our log-log fit of the published counts of Trenev *et al.* [29]). GeoVac uses raw Jordan–Wigner; Gaussian uses 2-qubit reduction (conservative for GeoVac).

$n_{\max}$	Q	GeoVac	Gaussian (interp.)	Ratio
1	14	22	1 126	51
2	70	1 954	580 688	297
3	196	99 247	31 457 102	317

†**H_2O 1-norm.** The raw H_2O 1-norms are dominated by the Z^2-scaled Phillips–Kleinman barrier at $Z_{\text{eff}} = 6$. PK contributes 98.7% of the total 1-norm: the single PK diagonal element on the O-side 1s orbital is $\sim 2{,}387$ Ha, replicated across four O-side valence blocks (2 bond + 2 lone pair)—measured 2026-08-30 as 2386.769 Ha exactly four times. Excluding PK, the electronic 1-norm is $\lambda_{\text{elec}} = 250$ Ha ($n_{\max} = 1$) and 372 Ha ($n_{\max} = 2$), comparable to BeH_2's total 1-norm. For LiH ($Z = 3$), PK contributes only 11.7% of the 1-norm ($\lambda_{\text{elec}} = 34$ vs. $\lambda_{\text{total}} = 39$ at $Q = 30$). The Z^2 scaling of PK parameters inflates the raw 1-norm for heavier atoms.

PK classical partitioning. Because PK is a one-body operator, its energy contribution $E_{\text{PK}} = \text{Tr}(h^{(1)}_{\text{PK}} \cdot \gamma)$ can be computed classically from the 1-RDM γ measured during VQE, with zero additional quantum circuits. The operator decomposition $H_{\text{full}} = H_{\text{elec}} + H_{\text{PK}}$ is verified to machine precision ($< 10^{-12}$) for all three composed systems, and the algebraic identity $E_{\text{full}} = \langle\psi|H_{\text{elec}}|\psi\rangle + \langle\psi|H_{\text{PK}}|\psi\rangle$ holds with residual $< 10^{-13}$ Ha. This quantum–classical partitioning reduces the H_2O quantum Hamiltonian 1-norm from 28,061.9 Ha to 372.5 Ha (75.3×), with no approximation. The partitioned electronic 1-norms across the composed systems all sit far below their with-PK values: LiH 34.0 Ha, BeH_2 68.8 Ha, H_2O 372.5 Ha (re-measured 2026-08-30; the retired figures were 359 / 78× / 33 / 66).

The commutator-based bound (Sec. III H) provides an additional 4–5× step reduction for composed systems at $n_{\max} \geq 2$, consistent with the single-geometry tightening ratio.

D. Comparison with Gaussian bases

We compare the composed GeoVac Pauli counts against the published Gaussian data of Trenev *et al.* [29] (Table 5, Appendix B) for LiH and H_2O (Table IX).

For LiH, the three published Gaussian data points (STO-3G at $Q = 10$, 6-31G at $Q = 20$, cc-pVDZ at $Q = 36$) yield a fitted exponent $\alpha_G = 4.25$ ($R^2 = 0.999$). For H_2O, the corresponding points ($Q = 12, 24, 46$) give $\alpha_G = 3.92$ ($R^2 > 0.999$).

Table XII shows the equal-qubit comparison for H_2O, using our fitted Gaussian power law for interpolation.

At $Q = 196$, the composed GeoVac Hamiltonian for H_2O requires $\sim 320\times$ fewer Pauli terms than the Gaussian power-law interpolation at the same qubit count, and $\sim 50\times$ at the smallest basis ($Q = 14$). (Corrected 2026-08-29: the retired pair-diagonal counts gave 51×–1,712× here.) The multiplier grows from $Q = 14$ to $Q = 196$ because the exact-rule exponent (3.17 over this range) still sits below the Gaussian 3.92; the paper's decisive comparison, however, is the *system-size* axis, where the composed count is exactly linear in Q at fixed basis while the Gaussian count is not.

For direct (non-interpolated) comparison: at $Q = 70$ ($n_{\max} = 2$), GeoVac produces 1 954 Pauli terms for H_2O, while the recomputed Gaussian STO-3G value is 551 terms at $Q = 12$. At $Q = 196$ ($n_{\max} = 3$), GeoVac's 99 247 terms sits just below the published Trenev Gaussian cc-pVDZ count of 107 382 at $Q = 46$ (0.92×; under the retired rule this read "far below" at 5.8×). These comparisons at *different* qubit counts illustrate that GeoVac achieves comparable term counts while encoding substantially more orbital degrees of freedom.

E. Cross-block ERI bound

The pseudopotential approximation sets cross-block valence ERIs to zero. To bound the impact of this approximation, we count the number of cross-block index quadruplets satisfying the Gaunt selection rules.

For two valence blocks with identical angular quantum numbers, the cross-block ERI count equals the same-block count exactly, because both blocks use the same hydrogenic orbital set. Including cross-block ERIs would at most double the ERI Pauli contribution. Since N_{ERI} dominates N_{h1} at all basis sizes, this would at most double the total Pauli count—shifting the exponent by at most $\log 2/\log Q_{\max} \approx 0.13$, keeping it well below 3.0 even in the worst case.

Empirical verification (Track CB, v2.0.37, measured under the retired pair-diagonal rule — the ratio, not the absolute counts, is the claim): for LiH at $n_{\max} = 2$ ($Q = 30$), the cross-block Coulomb ERI count was 130 versus 195 within-block (ratio 2/3), and the Pauli count with cross-block ERIs included was 854 (2.56× the pair-diagonal composed 334; the exact-rule composed count is 838). The $\leq 2\times$ bound is slightly exceeded because the theoretical bound counts only same-angular-set pairs, while the core block ($Z = 3$, max $l = 1$) has different radial extent than valence ($Z_{\text{eff}} = 1$), producing additional nonzero quadruplets. The 1-norm increases from 37.3 to 85.7 Ha (2.30×), and QWC groups from 21 to 88 (4.19×).

Importantly, simply adding cross-block ERIs while removing the PK pseudopotential does *not* improve accuracy: a 4-electron FCI test gives 29% error (coupled) versus 15% (composed with PK) versus 10.9% (no PK, no cross-block). The root cause is that the composed orbital basis uses different effective charges per block, so cross-block ERIs add electron–electron repulsion without the balancing cross-center nuclear attraction (one-body terms where each orbital feels all nuclei). This confirms that the pseudopotential is not redundant in second quantization for the composed basis: PK encodes a one-body effective potential that approximates both core–valence repulsion and orthogonality enforcement, and replacing it with explicit two-body terms is incomplete without also adding two-center one-body terms. The block-diagonal ERI structure thus remains the correct operating point for the composed architecture.

Note added (v2.0.42). Paper 19 [30] resolves the incompleteness identified above. The "balanced coupled" Hamiltonian adds both cross-block ERIs *and* cross-center nuclear attraction integrals $\langle\psi^A_{nlm}|V_B|\psi^A_{n'l'm'}\rangle$, computed analytically via incomplete gamma functions (machine precision). The multipole expansion of V_{ne} terminates exactly at $L_{\max} = 2l_{\max}$ by Gaunt selection rules. The resulting Hamiltonian produces the only bound LiH configuration in 4-electron FCI (1.8% energy error vs PK's 15.0%) and achieves 0.20% energy error at $n_{\max} = 3$ with analytical integrals. The resource tradeoff is regime-dependent: for VQE/NISQ (Pauli-count dominated), composed with PK remains cheaper; for QPE (1-norm dominated), balanced coupled eliminates PK entirely, which is decisive for heavy atoms where PK's Z^2 barrier dominates the 1-norm. See Sec. IV H for the resource census.

F. Limitations of the composed-system analysis

The comparison in this section quantifies quantum *simulation cost* (Pauli terms, measurement groups, 1-norm), not basis-set quality. GeoVac uses hydrogenic spin-orbitals in natural geometry blocks while Gaussian methods use contracted GTOs; the physical content per qubit differs. The metrics that determine quantum simulation cost—Pauli term count, 1-norm, and qubit count—are independent of the classical PES accuracy. Several approximations enter the composed-system results, and we document them below, distinguishing those that affect simulation cost from those that affect only energy accuracy:

1. **Cross-block ERIs set to zero:** The pseudopotential approximation eliminates all cross-block two-electron integrals. Selection-rule counting (Sec. IV E) bounds the impact at $\leq 2\times$ the reported Pauli counts. *Resolved:* Paper 19 [30] includes all cross-block ERIs with Gaunt selection rules preserved across block boundaries.

2. **Cross-center nuclear attraction omitted:** One-body cross-center terms (e.g., H orbital feeling the Li nucleus) are not included. These affect N_{h1} but not ERI scaling; at typical bond lengths the cross-center overlap decays exponentially. *Resolved:* Paper 19 computes these integrals analytically via incomplete gamma functions. 33 terms at $n_{\max} = 2$, 168 at $n_{\max} = 3$.

3. **Phillips–Kleinman parameters Z^2-scaled:** The repulsive barrier coefficients are Z^2-scaled from the Li^{2+} core, not computed ab initio for each system. This affects energy accuracy but not the Pauli count scaling.

4. **Valence Z_{eff} is constant:** The effective nuclear charge seen by valence electrons is position-independent, not r-dependent. A self-consistent $Z_{\mathrm{eff}}(r)$ would change energies but not the orbital block structure.

5. **Gaussian comparison uses 2-qubit reduction:** The published Trenev Gaussian Pauli counts use a 2-qubit reduction (particle number and spin symmetry); GeoVac does not. This makes the comparison conservative for GeoVac.

6. **R_{eq} error measures PES shape, not single-point quality:** The reported R_{eq} errors (5.3% for LiH, 11.7% for BeH_2, 26% for H_2O) reflect the equilibrium distance of the composed PES, not the energy accuracy at any particular geometry. For quantum simulation at a fixed nuclear configuration—the standard use case in fault-tolerant resource estimation [7, 32]—the relevant comparison is simulation cost at the same geometry and qubit count.

7. **Valence-valence coupling not captured:** The block-diagonal approximation eliminates inter-bond-pair electron interactions. For BeH_2, this produces a qualitatively correct PES (bound state at $R_{\mathrm{eq}} = 3.31$ bohr) but quantitatively poor accuracy (32% error vs. experiment). Two correction approaches were tested: (i) Z_{eff} partitioning (dividing screened charge among bonds) worsened R_{eq} to 103% error; (ii) classical point-charge inter-bond repulsion moved R_{eq} in the wrong direction. The root cause is that bond pairs sharing a nucleus require orbital-level coupling (exchange and orthogonalization), not scalar energy corrections. This limitation applies to the *accuracy* of composed polyatomic PES, not to the Pauli term scaling, which remains valid as a lower bound. *Partially resolved:* Paper 19 includes cross-block ERIs and cross-center V_{ne} between all block pairs. BeH_2 4-electron FCI improves from 27.4% (PK) to 10.7% (balanced).

G. Transcorrelated variant

The electron-electron cusp ($1/r_{12}$ singularity) limits basis convergence at each level of the natural geometry hierarchy (Paper 18, Sec. II). The transcorrelated (TC) method removes this singularity via a similarity transformation $H_{\mathrm{TC}} = e^{-J} H\, e^{J}$ with Jastrow factor $J = \frac{1}{2} r_{12}$. For the composed qubit pipeline, the TC transformation

TABLE XIII. Standard vs. transcorrelated (TC) composed Hamiltonians at $n_{\max} = 2$. "Pauli ratio" is TC/Standard Pauli term count. "1-norm ratio (elec.)" compares electronic-only 1-norms with PK classically partitioned. The TC constant factor of 1.616× is uniform across all molecules. *Re-measured 2026-08-30* under the exact global-M_L rule, discharging the vintage note that stood here: the retired pair-diagonal row read 334/562, 556/934, 778/1,307 with 1-norm ratios 1.09/1.16/1.00, and the constant was 1.68×. Pauli counts exclude the identity term.

Molecule	Q	Std Pauli	TC Pauli	1-norm ratio
LiH	30	837	1 353	1.28
BeH_2	50	1 395	2 255	1.29
H_2O	70	1 953	3 157	1.25

replaces the standard Coulomb ERI with a gradient operator $G = -\frac{1}{2}\,\hat{r}_{12} \cdot (\nabla_1 - \nabla_2)$, while leaving one-body integrals unchanged. A constant shift of $-\frac{1}{4}$ per electron pair arises from the Baker-Campbell-Hausdorff second-order term $\frac{1}{2}|\nabla J|^2$. The BCH expansion terminates exactly at second order for two-electron blocks, and three-body terms vanish between composed blocks (no shared electrons).

The TC Hamiltonian is non-Hermitian: the gradient operator G breaks the chemist symmetry $\langle pq|rs\rangle \neq \langle rs|pq\rangle$, so the Jordan-Wigner transformation produces additional Pauli terms from the antisymmetric part of the ERI. However, the Gaunt angular selection rules—which determine *which* index quadruplets are nonzero—are preserved, because $J = \frac{1}{2}r_{12}$ is rotationally invariant. The TC transformation densifies the radial integrals within each allowed angular channel but does not create new angular channels. The quantum-algorithm cost of this non-Hermiticity—and whether it is worth avoiding—is analysed in Sec. IV G 1.

Table XIII compares standard and TC composed Hamiltonians at the production operating point ($n_{\max} = 2$).

The Pauli ratio is 1.616× for all three molecules, identical to three decimals — so the cross-molecule uniformity survived the exact-rule correction, with the constant moving from the retired 1.68×. *The scope caveat from the 2026-08-29 vintage note is retained, not retired with the numbers:* this is a fixed-basis ($n_{\max} = 2$) cross-molecule measurement, so it establishes uniformity *across molecules* and not that the TC overhead leaves the basis-axis exponent unchanged — the same single-point-fallacy risk the correction exposed elsewhere. Electronic-only 1-norm overhead is a tight 25–29% (1.283 / 1.293 / 1.253× for LiH / BeH_2 / H_2O). The retired figures were 9–16% with H_2O at exactly 1.00×; that coincidence, and the PK-dominance explanation built on it, do not survive re-measurement.

For He at the atomic level, the TC variant eliminates a basis divergence: standard hydrogenic FCI error grows from 5.3% ($n_{\max} = 1$) to 8.2% ($n_{\max} = 3$), while TC error converges to a 3.3–3.6% plateau. The plateau persists even with the full gradient operator (Track BX-4); the radial-only restriction accounts for only 0.01 pp of the ∼3.3% floor, which is intrinsic to the hydrogenic basis at this truncation level.

Angular gradient benchmark (Track BX-4). The full angular gradient term of G was implemented and benchmarked for $l > 0$ orbitals. For He, including the angular gradient reduces the FCI error by 0.010–0.014 pp: radial-only gives 3.621% at $n_{\max} = 2$ and 3.639% at $n_{\max} = 3$; full TC gives 3.611% and 3.625%, respectively. The accuracy gain is negligible. The Pauli cost, however, is substantial: 188 → 500 terms at $n_{\max} = 2$, and 1,492 → 3,449 ERIs at $n_{\max} = 3$. *The composed-pipeline angular-gradient multipliers are withdrawn* (2026-08-30): they read 562 → 1,498 (LiH), 936 → 2,496 (BeH_2), 1,310 → 3,494 (H_2O), a uniform 2.67× and a 4.49× total, but the operator producing them is defective. Every entry the angular gradient adds to the composed ERI tensor violates L_z: measured 78/78 added entries for LiH, against a control in which the plain composed and radial-only TC tensors have *zero* violations under the same test. A two-body Coulomb-type operator cannot add L_z-violating support, so those entries are spurious and the multipliers measure the defect as much as the operator. They are withdrawn rather than re-measured, because re-measuring a broken operator produces a number that has to be disclaimed in the same sentence. The angular gradient remains a *negative result for quantum efficiency* — the verdict is now over-determined, since the accuracy gain was already negligible (0.010–0.014 pp) and the added support is unphysical. The radial-only TC Hamiltonian (`include_angular=False`) remains the default for the composed pipeline.

A PES scan for LiH confirms that both the standard and TC Pauli counts are R-independent (837 and 1,353 at all sampled bond lengths under the exact rule; the scan was run at the retired 334 and 562, and R-independence is a support property that the value correction does not disturb), consistent with the selection-rule origin of the sparsity.

1. Quantum-algorithm cost of the non-Hermitian form

The similarity transformation makes H_{TC} non-Hermitian, placing it outside the Hermitian resource model used above (Rayleigh-quotient VQE, real-time Trotter, and textbook QPE all assume a Hermitian generator). We characterise this cost on an atomic Coulomb–Sturmian realisation and reach two conclusions: the non-Hermitian form is cheap to run, and the naive Hermitisation that would avoid it sacrifices accuracy.

The non-Hermiticity is localised and benign. For a three-electron reference (Li $1s^22s$) the electron–electron cusp activates a genuine three-body operator, which we contract to an effective two-body operator using the reference one-particle density matrix (the xTC scheme [38]).

This contracted operator is Hermitian to machine precision ($\|A-A^\dagger\| \leq 3\times10^{-18}$ for an s reference, 3×10^{-17} for an open-p reference); the only non-Hermitian term is the generic convective operator G, present in any transcorrelated Hamiltonian. For the full operator the spectrum is entirely real, the ground state real, and the departure from normality small (0.015 for Li, 0.069 for He). The cost of eigenvalue estimation for a non-normal block-encoded operator is governed by the eigenvector condition number κ_V (the Bauer–Fike factor), which enters *linearly* in the optimal quantum eigenvalue-processing route [39]. We measure $\kappa_V = 1.21$ for Li (grid-stable) and $\kappa_V = O(1)$, $\lesssim 2$, for He, so the penalty is a small constant factor rather than a change of scaling; the Pauli 1-norm is barely affected ($\lambda_{\mathrm{TC}}/\lambda_{\mathrm{std}} \leq 1.13$). Independent work on transcorrelated electronic Hamiltonians in Gaussian bases reaches the same conclusion ($\kappa_V \approx$ 1–3, reduced further by xTC) [40]. The one genuine overhead is the Pauli *count*: the non-symmetric G tensor roughly doubles it ($117 \to 249$ on the atomic s-only engine).

Hermitisation by symmetrisation is a false economy. The symmetrised operator $\frac{1}{2}(H_{\mathrm{TC}} + H_{\mathrm{TC}}^\dagger)$ is attractive on cost—it restores the plain Pauli count, lowers the 1-norm slightly below plain (0.98×), and is Hermitian ($\kappa_V = 1$)—but it is a different, less accurate operator. Discarding the anti-Hermitian part of G shifts the energy downward past the exact value, and the overshoot *worsens* as the basis improves: at the largest basis tested the genuine transcorrelated energy lies +5.4 mHa (He) and +7.7 mHa (Li) above the exact non-relativistic energy and approaches it from the variationally correct side, whereas the symmetrised energy lies −6.6 and −21.4 mHa *below* exact. No stationary Jastrow range anchors the symmetrised operator (E_{TC} is monotone in the geminal exponent). We therefore run the operator in its native non-Hermitian form—affordable by the $\kappa_V = O(1)$ result above—rather than symmetrising. Deleting G altogether is not an alternative (the remaining two-body term then over-binds by $\sim$ 0.6 Ha), and folding to $H_{\mathrm{TC}}^\dagger H_{\mathrm{TC}}$ is excluded because it returns the smallest singular value, not the ground eigenvalue, and squares the conditioning. These measurements use the atomic s-only Coulomb–Sturmian engine (He, Li) with a single spin-independent geminal; the robust content is the *ordering* (native non-Hermitian more accurate than symmetrised, both correcting the cusp relative to plain), not an absolute accuracy figure for this minimal basis.

2. Angular sparsity survives the cusp correction (atomic xTC)

The cost analysis above establishes that the non-Hermitian TC operator is affordable. The complementary question is whether it is still *sparse*: when the electron–electron cusp activates a genuine three-body operator and we contract it to an effective two-body operator against the reference density (the xTC scheme [38]), does the contracted operator inherit the angular Gaunt sparsity of the plain Coulomb operator (Paper 22), or does the three-body → two-body contraction fill in the zero angular blocks? The question is sharp because the transcorrelated vertex is *non-abelian*: a leg harmonic and a correlator harmonic Clebsch–Gordan couple to a *multiplet* of resultants, not a single one, which is exactly why the three-body operator does not collapse to two-body the way the abelian plane-wave TC does.

TABLE XIV. Exact angular fill-in of the xTC-contracted two-body operator relative to the plain Coulomb operator, swept across the monoatomic library. "Fill-in" counts blocks nonzero under xTC but zero under Coulomb; fill-out is zero in every case, so zero fill-in means the support (hence the Jordan–Wigner Pauli term set) is preserved exactly. Spherical reference: any closed, half-filled high-spin, or s-open shell (Unsold; e.g. Be $1s^22s^2$, N $2p^3$, Mn $3d^5$); open-p: C/O ($2p^2/2p^4$); open-d: Sc/Ti-type ($3d^1/3d^2$). Counts are exact integers (angular support is radial-independent).

Basis	spherical	open-p	open-d	Coulomb density
$s+p$	0	0	0	14.8%
$s+p+d$	0	4	4	8.5%
$s+p+d+f$	0	96	96	6.1%
$s+p+d+f+g$	0	264	268	4.8%

The answer is settled at the level of angular *support*—which orbital quadruplets are nonzero—which is radial-independent and therefore exact (integer block counts, no quadrature). Table XIV gives the fill-in (blocks xTC makes nonzero where Coulomb was zero) for spherical, open-p, and open-d references across increasingly rich bases, swept over the whole monoatomic library $Z = 1$–56.

The fill-in is governed by a single exact rule: it is *zero at every basis*, bit-exactly, precisely when the reference density is spherical. By Unsöld's theorem the spherical shells are the closed subshells, the half-filled high-spin subshells, and the s-open shells—so the zero-fill-in guarantee covers not only the framework's headline molecules H_2, LiH, BeH_2 (all s-referenced, Be is $1s^22s^2$) but also, e.g., nitrogen ($2p^3$) and chromium/manganese ($3d^5$). Across the $Z = 1$–56 library this is 29 of the 56 atoms; the remaining 27 split into 16 open-p and 11 open-d references. The mechanism is the same monopole collapse in every spherical case: a spherical reference density carries only the $L' = 0$ multipole (Unsöld), so contracting a correlator leg against it forces the leg to the monopole and collapses the non-abelian vertex to a single Coulomb-like multipole with identical angular support. Because the support is unchanged (and no block is removed either), the Jordan–Wigner Pauli term count is preserved exactly; on the $s+p$ radial engine (Li reference) the plain and xTC operators both carry 279 Pauli terms, and the two-body 1-norm is *lowered* to 0.76–0.91× plain across geminal widths $\gamma \in [0.6, 1.5]$. So xTC handles the cusp without inflating the qubit cost.

A non-spherical open-p reference (partially filled $2p$

that is not half-filled: C, O, F, and the oxygen of H_2O—but *not* N, whose $2p^3$ is half-filled and therefore spherical) carries the additional $L' = 2$ density multipole, which lets the vertex bridge an even-multipole gap between external pairs. The fill-in is still zero through $s + p$ (at $l \leq 1$ every same-parity pair already contains the monopole, so Coulomb misses nothing), and only a small, structured fill-in appears once d orbitals enter—4 blocks at $s+p+d$, 96 at $s + p + d + f$, 264 at $s + p + d + f + g$. An open-d reference (the transition metals whose d-shell is neither empty, half, nor full) additionally carries the $L' = 4$ multipole; this makes *no* difference through $s + p + d + f$ (identical 0/4/96) and first bites at $s + p + d + f + g$, where the open-d fill-in is 268 against the open-p 264—a genuine but tiny $L' = 4$ effect (+4 blocks out of 390,625). In every case the xTC density stays within 2.5% relative of the Coulomb density (both fall from 14.8% at $s+p$ to 4.8% at $s+p+d+f+g$), so the operator does not densify across the physical range; the honest difference from the spherical case is that the preservation is no longer bit-exact and the fill-in fraction rises slowly with $l_{\max}$. The whole pattern is a periodicity statement (Paper 16): the fill-in class is fixed by the open shell's angular momentum, so it repeats down each group of the periodic table.

This is an atomic property. The angular sparsity that xTC preserves is the single-center l-selection of Paper 22, and that sparsity does not survive to two centers: a two-center Coulomb operator already sits at 100% of its m-conservation limit (LiH) against the atom's 55%, so a molecule has lost the rich l-sparsity *before* xTC is applied. For a molecule xTC therefore preserves only the m-conservation sparsity plus a modest 1-norm reduction; the combined "cusp corrected and angular sparsity preserved" result is strongest for atoms. As throughout this section, these are single-common-k Coulomb–Sturmian measurements with a single spin-independent geminal on a classical non-Hermitian proof of concept—structural (support and resource) results, not absolute accuracy claims for the minimal basis.

H. Balanced coupled variant

Paper 19 [30] presents a PK-free alternative: the balanced coupled Hamiltonian includes all four interaction types—within-block h_1 and ERI, cross-block ERI, and cross-center nuclear attraction $\langle\psi^A|V_B|\psi^A\rangle$—with no pseudopotential. Cross-center V_{ne} integrals are computed analytically via incomplete gamma functions; the multipole expansion terminates exactly at $L_{\max} = 2l_{\max}$ by Gaunt selection rules.

Table XV compares composed and balanced encodings at $n_{\max} = 2$.

The Pauli ratio grows with block count (3.26× to 10.1×), but the 1-norm ratio favors balanced for BeH_2 (0.86× composed with PK in-Hamiltonian, the QPE-relevant convention; live builders, re-measured 2026-09-01 at BeH_2's own bond length $R = 2.502$ bohr — 323.4 vs. 374.9 Ha, both identity-included, since the composed-with-PK total is only available that way) where PK's Z^2-scaled barrier is costly. For H_2O, the balanced 1-norm (1,612.0 Ha) is 17.4× lower than the full composed 1-norm with PK (28,061.9 Ha). Composed and balanced are thus complementary encoding strategies: composed is cheaper for VQE (Pauli-count dominated), balanced is cheaper for QPE on heavy atoms (1-norm dominated, PK eliminated). The balanced coupled encoding is available via `build_balanced_hamiltonian()` in the production codebase.

TABLE XV. Composed vs. balanced coupled resource census at $n_{\max} = 2$. *Pauli counts and λ both exclude the identity term*, on both sides — re-synced 2026-08-30, when the balanced columns were found to be identity-*included* and the composed λ column identity-included, under a caption declaring otherwise. Composed λ values are electronic-only 1-norms (PK classically partitioned via 1-RDM; the partitioning machinery is implemented in `geovac/pk_partitioning.py` per CLAUDE.md § 7). Balanced λ values include cross-center V_{ne} (no PK). Paper 20's resource table reports the same systems with the identity term *in*, which is why its cells are larger.

System	Q	Pauli terms		λ (Ha)	
		Comp.	Bal.	Comp.	Bal.
LiH	30	837	2,725	27.7	75.2
BeH_2	50	1,395	8,867	54.3	286.9
H_2O	70	1,953	19,741	186.8	1,439.8

I. Discussion of composed results

The composed-geometry approach preserves and improves the structural sparsity of single-geometry GeoVac Hamiltonians. Three factors drive this:

1. **Block-diagonal ERIs:** The pseudopotential approximation eliminates all cross-block two-electron integrals, reducing the ERI count from $O(M^4)$ to $O(\sum_b M_b^4)$.
2. **Angular selection rules preserved:** Each block retains the Gaunt integral sparsity from the single-center case, with ERI density decreasing as $\sim 1/M_b$ within each block.
3. **Composition is additive:** Adding a new atomic center adds a new ERI block but does not create cross-block entries.

The universality of the within-molecule exponent (3.17, identical across all eight systems tested, spanning 1–7 orbital blocks and three periodic-table rows) is a consequence of an exact factorization: the composed Pauli count is $N_{\text{Pauli}} = c(n_{\max}) \times Q$ with molecule-independent rational coefficients $c(1) = 3/2$, $c(2) = 279/10 = 27.90$, $c(3) = 7089/14 \approx 506.36$, so every molecule's (Q, N_{Pauli})

TABLE XVI. Balanced coupled resource metrics for second-row molecules at $n_{\max} = 2$, with composed metrics for comparison. All molecules use [Ne] frozen cores with Clementi–Raimondi orbital exponents. Pauli counts include the identity term; λ_{ni} excludes it. *Each row is evaluated at its own equilibrium bond length R, stated in the table.* The geometry is free-side calibration input, so the two columns that depend on it are only meaningful once it is named: λ_{ni} sums coefficient magnitudes, and the QWC count comes from a greedy grouper whose path depends on those magnitudes. The Pauli count does *not* depend on R — it is fixed by the angular selection rules alone, and is bit-identical across geometries (`tests/test_paper20_geometry_independence.py`). (Corrected 2026-08-31: the λ_{ni} and QWC columns had been computed at $R = 3.015$ bohr — LiH's bond length — for every molecule, because the balanced builder defaulted to that value and ignored the spec's geometry. Retired cells: λ_{ni} 20.6/110.5/869.7/879.6/895.3/914.1 and QWC 1,282/1,477/1,551 for $H_2S/PH_3/SiH_4$.)

Molecule	Q	R (bohr)	Pauli terms		λ_{ni} (Ha)	QWC
			Comp.	Bal.	Bal.	Bal.
NaH	20	3.566	559	575	19.6	69
MgH_2	40	3.261	1,117	4,861	111.8	903
HCl	50	2.409	1,396	9,824	866.1	1,173
H_2S	60	2.534	1,675	13,863	873.6	1,264
PH_3	70	2.680	1,954	18,854	889.3	1,427
SiH_4	80	2.800	2,233	24,745	909.2	1,566

sweep is a horizontal translate of every other's on the log–log plane. Across the composed molecules at $n_{\max} = 2$ the count is exactly linear in Q: $N_{\mathrm{Pauli}} = 27.90 \times Q$, confirming that the linear system-size scaling extends to arbitrary polyatomic systems via the fiber bundle decomposition [23]. (Corrected 2026-08-29 from the retired pair-diagonal $11.10 \times Q$ and $Q^{2.5}$; under the retired rule H_2 showed an anomalous exponent 3.13 vs. the molecules' 2.50–2.52 — the exact rule dissolves the anomaly.)

J. Second-row molecules via frozen-core composition

The composed architecture extends naturally to second-row atoms ($Z = 11$–18) by replacing the Level 3-solved He-like core with an analytical frozen core. Each [Ne]-like core ($1s^2\,2s^2\,2p^6$, 10 electrons) is described by hydrogenic wavefunctions with Clementi–Raimondi orbital exponents, producing a smooth $Z_{\mathrm{eff}}(r)$ screening function (Z at the origin, $Z - 10$ at infinity). The frozen-core energy enters the qubit Hamiltonian only through the identity operator and does not affect quantum measurement cost.

Table XVI lists balanced coupled resource metrics for six second-row molecules at $n_{\max} = 2$.

The within-molecule Pauli scaling of all four second-row molecules (NaH, MgH_2, HCl, SiH_4) yields an exponent of 3.17, identical to the first-row result to all reported digits (the $N_{\mathrm{Pauli}} = c(n_{\max})\,Q$ factorization). This confirms that the scaling is universal across periodic table rows—driven by Gaunt selection rules that are independent of the core treatment.

Because second-row molecules use frozen cores (no encoded core block), they require fewer qubits than their first-row isostructural analogues: NaH uses $Q = 20$ versus LiH's $Q = 30$. Within a given row, however, the Pauli count depends only on the orbital block topology and $n_{\max}$. The key structural observation is that the Pauli *scaling exponent* (3.17) is universal across both rows, driven by Gaunt selection rules that are independent of nuclear charge.

For 1-norm comparisons, we report λ_{ni}—the non-identity component of the Pauli 1-norm—which excludes the frozen-core energy offset and nuclear repulsion (classical constants that do not contribute to quantum measurement cost or Trotter error). For NaH at its equilibrium $R = 3.566$ bohr, the identity term accounts for the bulk of the total 1-norm (191.9 Ha total vs. 19.6 Ha non-identity); the simulation-relevant $\lambda_{\mathrm{ni}} = 19.6$ Ha is comparable to first-row LiH ($\lambda_{\mathrm{ni}} = 27.7$ Ha at its own $R = 3.015$ bohr).

The framework extends further to third-row main-group atoms ($Z = 19$–20 and $Z = 31$–36) via [Ar] (18-electron) and [Ar]$3d^{10}$ (28-electron) frozen cores with Clementi–Raimondi orbital exponents. Six additional molecules—KH, CaH_2, GeH_4, AsH_3, H_2Se, HBr—bringing the single- and two-center library to 18 molecules across three periodic table rows (extended to 28 with multi-center systems in Sec. IV K). The universal 3.17 scaling exponent is maintained.

A strong structural invariance is confirmed: frozen-core molecules with the same block topology produce *identical* Pauli counts. For example, NaH and KH both yield 559 composed / 575 balanced Pauli terms at $Q = 20$, despite different core types ([Ne] vs. [Ar]) (Table XVI; re-measured 2026-08-30 under the exact global-M_L rule, which raised both counts from the retired pair-diagonal 223/239 while leaving the invariance itself intact). This invariance holds exactly across all six isostructural pairs spanning rows 2 and 3.

Basis convergence at $n_{\max} = 3$ (requiring $\ell = 2$ rotation matrices, now implemented via the Wigner D-matrix) produces NaH Hamiltonians at $Q = 56$ with 5,349 balanced Pauli terms. Two-electron FCI gives a 0.40 Ha energy improvement over $n_{\max} = 2$ at the reference geometry, confirming variational convergence. However, the potential energy surface retains the overattraction observed at $n_{\max} = 2$—no equilibrium geometry is found—indicating that PES shape accuracy is limited by the frozen-core approximation, not the angular basis.

K. Multi-center molecules

The composed architecture handles molecules with more than two nuclei through the same block-diagonal construction: each electron pair occupies an independent

TABLE XVII. Composed (block-diagonal) Pauli term counts for multi-center molecules at $n_{\max} = 2$. N_e: total electrons; N_b: orbital blocks; $Q = 2M$: qubits. Frozen cores are used where applicable (Na: [Ne], Cl: [Ne]). Counts are non-identity terms, measured from the shipping library; each is $27.90 \times Q$ exactly, the block coefficient of Sec. VI A 1. Corrected 2026-08-29: the retired pair-diagonal rule gave $11.10 \times Q + 1$.

Molecule	N_e	N_b	Q	Pauli terms
LiF	12	6	70	1,953
CO	14	7	100	2,790
N_2	14	7	100	2,790
F_2	18	9	100	2,790
NaCl	8	4	50	1,395

orbital block, and cross-block ERIs are zero by the pseudopotential approximation. Table XVII reports composed Pauli counts for the five multi-center systems in the shipping library at $n_{\max} = 2$.

The most striking feature of Table XVII is *isostructural invariance*: CO and N_2 produce identical Pauli counts (2,790) despite different nuclear charges, confirming that the invariance observed for frozen-core pairs (Sec. IV J) extends to multi-center topologies. Similarly, CO, N_2, and F_2 all yield 2,790 terms at $Q = 100$. The Pauli count is determined entirely by the qubit count Q, not by the number of blocks, the block structure, or the atomic species.

Across the 35 composed molecules in the library—spanning $Z = 1$–56 and including the transition metal hydrides (Sec. IV L)—the composed Pauli count obeys

$$N_{\text{Pauli}} = 27.90 \times Q, \tag{17}$$

where N_{Pauli} excludes the identity term. The coefficient 27.90 (= 279/10) is exact (not a fit) across all main-group molecules (corrected 2026-08-29; the retired pair-diagonal rule gave 11.10). This exact linearity in Q is a stronger result than the within-molecule basis scaling (Eq. 16), which is measured by varying $n_{\max}$ at fixed molecular topology. Equation (17) describes the *across-molecule* scaling at fixed $n_{\max} = 2$: adding more orbital blocks (more nuclei, more electron pairs) increases Q and N_{Pauli} in exact proportion, because each block contributes an independent set of Pauli terms with no cross-block coupling.

With these five multi-center systems, the GeoVac library spans 37 systems (35 composed molecules plus the He and H_2 reference atoms) across $Z = 1$–56 (H through Ba), including first-row (H_2 through CH_4), second-row (NaH through SiH_4), third-row (KH through HBr), multi-center (LiF, CO, N_2, F_2, NaCl), and the ten transition metal hydrides (ScH–ZnH; Sec. IV L). All are accessible via the `hamiltonian()` API with ecosystem export to Qiskit, OpenFermion, and PennyLane.

TABLE XVIII. ERI density and Pauli term counts for single orbital blocks of different angular momentum composition at $n_{\max} = 2$. M: spatial orbitals; $Q = 2M$: qubits.

Block type	M	Q	ERI density	Pauli	Pauli/Q
s-only ($l_{\max} = 0$)	3	6	100%	118	19.67
s+p ($l_{\max} = 1$)	9	18	15.7%	2,967	164.83
d-only ($l = 2$)	5	10	13.6%	343	34.30

L. d-Orbital blocks and transition metal hydrides

Higher angular momentum admits *more* multipole orders, not fewer: for d-orbital blocks ($l = 2$) the allowed orders are $k = 0, 2, 4$ (three values) against $k = 0, 1$ for s+p. *Corrected 2026-08-30:* this section previously opened “the Gaunt selection rules become *more* restrictive at higher angular momentum” and quoted 20% against 38.5% nonzero angular coefficients. Both were artifacts of the retired pair-diagonal rule, which deleted proportionally more of the d-block’s genuine m-changing couplings. Table XVIII quantifies the effect on ERI density and Pauli term counts for isolated blocks at fixed qubit counts.

The d-only block is *not* the sparsest. Its ERI density (13.6%) is close to the s+p block’s (15.7%), and per qubit it is the denser of the two: Pauli/Q = 34.30 against 27.90 for the s+p block at $n_{\max} = 2$ that dominates main-group molecules. Under the retired pair-diagonal rule these read 4.0% / 5.60 against 8.9% / 54.22, making the d-block look dramatically leaner. That was an artifact: higher angular momentum has *more* m-changing integrals to drop, so the pair-diagonal restriction deleted proportionally more of the d-block’s genuine couplings. The correction is consistent with the composed d-block coefficient (30.03 against main-group 27.90, Sec. III D): the d-block is denser in both measurements.

Transition metal hydrides are constructed via frozen-core composition. ScH ($Z = 21$) and TiH ($Z = 22$) carry an [Ar] frozen core (18 electrons) and valence blocks that include an isolated d-orbital block ($l_{\min} = 2$, $l_{\max} = 2$) alongside an s-orbital bond pair. At $n_{\max} = 2$, both molecules require $Q = 30$ qubits and produce 901 non-identity Pauli terms—more than LiH (837 at the same Q)—giving Pauli/Q = 30.03, slightly *above* the main-group coefficient of 27.90. Under the exact Coulomb rule d-blocks are marginally *denser* than s/p blocks, not sparser; the retired pair-diagonal figures (277 terms, Pauli/Q = 9.23, “below the universal 11.10”) inverted this ordering.

The atomic classifier now automates the full first transition series ($Z = 21$–30, type-F dispatch), and all ten hydrides (ScH–ZnH) ship in the 37-system library; the Hamiltonians remain resource-level results—they have not been benchmarked against FCI or Gaussian baselines. Nevertheless, the structural finding—that d-orbital blocks have a lower *pair-diagonal* Pauli/Q ratio than s/p blocks (Sec. III D)—is an artifact of the pair-diagonal

TABLE XIX. Nested vs. composed encoding for Be ($Z = 4$, 4 electrons, $n_{\max} = 2$). λ_{ni}: non-identity 1-norm.

Encoding	Q	Pauli	λ_{ni} (Ha)	FCI err
Nested (H-set)	10	112	18.95	4.9%
Composed + PK	12	115	121.35	1.0%

rule dropping more m-swap integrals at higher angular momentum (under the exact global-M_L rule the d-block is denser, not sparser), not a sparser exact selection rule.

M. Compact nested encoding

For single-center N-electron systems, a compact "nested" encoding places all electrons in a single orbital block on S^{3N-1}, using H-set coupled hyperspherical harmonics (Sec. III E) as the angular basis. This eliminates the core/valence decomposition and the PK pseudopotential entirely, reducing the qubit count by $\sim 3\times$ for 4-electron atoms.

Table XIX compares the nested and composed encodings for beryllium (4 electrons).

The nested encoding achieves comparable Pauli count (112 vs. 115) with 6.4× lower non-identity 1-norm (18.95 vs. 121.35 Ha) due to PK elimination. The accuracy tradeoff (4.9% vs. 1.0% FCI error) reflects the single-center basis truncation at $n_{\max} = 2$.

For molecules, the nested encoding requires per-pair effective nuclear charges (heterogeneous basis), which introduces Löwdin orthogonalization costs that destroy Gaunt-level sparsity: LiH with heterogeneous nested encoding produces 1,711 Pauli terms (measured under the retired pair-diagonal rule, vs. 120 uncoupled or 334 composed under that same rule; the exact-rule composed count is 838 — the Löwdin-inflation comparison is rule-consistent as quoted). The composed architecture's block-diagonal factorization avoids this by maintaining independent orthonormal bases at each length scale.

The nested encoding is most advantageous for atomic systems and single-point molecular calculations where qubit count is the binding hardware constraint.

V. SPINOR COMPOSED ENCODING

The composed qubit Hamiltonians of §IV use a spinless orbital basis with a trivial factor-of-two spin doubling at Jordan–Wigner encoding. This section reports the *spinful* extension: one qubit per Dirac label (n, κ, m_j) with the angular structure provided by Szmytkowski's spinor spherical harmonics and the spin-orbit coupling provided by the Breit–Pauli leading-order reduction of the Dirac equation. The construction is an engineering upgrade of §IV's composed pipeline, not a new Dirac Fock projection: the Schrödinger Fock map on S^3 does not lift to the Dirac sector (a $\mathbb{Z}_2$-graded superalgebra obstruction), so the spinor basis rides on the existing scalar S^3 graph with relativistic coefficients added on top. All matrix elements are closed-form in (κ, m_j)-native labels: no numerical quadrature enters the angular layer, and the radial layer reuses the exact-rational R^k evaluator from `hypergeometric_slater.py`.

A. Resource metrics for LiH, BeH, CaH

Three single-bond hydrides, spanning $Z \in \{3, 4, 20\}$ and covering both first-row (LiH, BeH) and the [Ar]-frozen-core third row (CaH). BeH is constructed as a one-bond reduction of BeH_2; CaH is the Ca analog of BeH with the third-row frozen-core decomposition implemented in `geovac/neon_core.py` (extended to [Ar] in v2.2.0; equivalently, the Z=20 entry in the atomic classifier). All three specs enumerate Dirac labels directly rather than spin-doubling a spatial-orbital basis; the resulting qubit count Q coincides with the scalar qubit count by construction because $2(l_{\max} + 1)^2$ Dirac labels at shell n equal $2 \cdot (l_{\max} + 1)^2$ spin-doubled scalar orbitals.

TABLE XX. Spinor composed encoding resource metrics for LiH, BeH, CaH at $n_{\max} \in \{1, 2, 3\}$. N_{Pauli} is the number of non-identity Jordan–Wigner Pauli strings; λ_{ni} is the non-identity 1-norm; "QWC" is the number of qubit-wise commuting groups (greedy grouping). "scalar" columns repeat the spinless encoding of §IV at the same spec topology for reference. Counts are non-identity Pauli terms; λ_{ni} is the corresponding 1-norm; QWC groups are greedy. *Re-measured 2026-08-30* under the corrected Condon–Shortley angular factor order (retired values: N_{Pauli} = 1,413 / 942 / 89,226 / 59,484 and λ = 40.59 / 143.96 / 18.68 / 218.78 / 413.90 / 125.36). The ordering was settled against a Clebsch–Gordan rotation of the verified scalar tensor; see `tests/test_paper14_spinor_eri_ordering.py`. The former "scalar" and "rel/scalar" columns have been *removed* rather than re-priced: measurement shows the relativistic spec evaluated by the spinless builder returns the same count as the relativistic builder in all nine rows, so those columns compared a quantity with itself and the "4.24× / 11.33×" ratios were not a spinor overhead. No scalar BeH or CaH spec exists to compare against; where a genuine scalar counterpart does exist (LiH at $n_{\max} = 2$) it is 837. Isostructural invariance survives and is cleaner than before: LiH and BeH agree exactly at every basis, and the QWC counts agree across all three molecules at each basis.

$n_{\max}$	Mol.	Q	$N^{\text{rel}}_{\text{Pauli}}$	$\lambda^{\text{rel}}_{\text{ni}}$ (Ha)	QWC$^{\text{rel}}$
1	LiH	6	9	10.15	1
1	BeH	6	9	63.44	1
1	CaH	4	6	2.03	1
2	LiH	30	1,501	39.53	142
2	BeH	30	1,501	142.52	142
2	CaH	20	998	17.91	142
3	LiH	84	90,114	208.79	11,938
3	BeH	84	90,114	400.01	11,948
3	CaH	56	60,060	119.35	11,925

Three observations on Table XX:

1. *The resource counts are isostructural.* At fixed $n_{\max}$, LiH and BeH give *identical* Pauli counts (9, 1,501, 90,114 at $n_{\max} = 1, 2, 3$) despite different nuclear charges and valence configurations, and the QWC counts agree across all three molecules at each basis (1, 142, $\approx$ 11.9k) even where Q differs (CaH at $Q = 20$ against $Q = 30$). This is what the general scaling law of §IV predicts: the composed count depends on Q and on the angular structure, not on nuclear charge. (Restated 2026-08-30. This observation previously reported a "rel/scalar Pauli ratio" of 1.00×/4.24×/11.33×; that ratio is withdrawn—its denominator turned out to be the same quantity as its numerator, so it measured no spinor overhead. The isostructural content, which is what the observation was for, survives and is stated here directly.)

2. *1-norm increases modestly; no order-of-magnitude inflation.* The relativistic non-identity 1-norm λ_{ni} is 10.15/39.53/208.79 Ha for LiH and 63.44/142.52/400.01 for BeH at $n_{\max} = 1, 2, 3$, and 2.03/17.91/119.35 for CaH (Table XX). No scalar-versus-relativistic ratio is quoted: the "scalar" comparator this observation formerly used was the same quantity as the relativistic one (see the table caption), so the former "$\sim$ 13% at $n_{\max} = 1, 2$, 20–30% at $n_{\max} = 3$" figures measured nothing. What can be said without that comparator is that the 1-norm stays modest in absolute terms. There is no order-of-magnitude inflation: spinor jj-coupling redistributes Coulomb matrix elements across angular channels, so the aggregate 1-norm stays the same order as the scalar encoding—the spinor Hamiltonian remains tractable for QPE resource estimation, where λ dominates query complexity [35].

3. *QWC grouping cost grows sharply with basis.* The relativistic LiH encoding produces 1, 142 and 11,938 QWC groups at $n_{\max} = 1, 2, 3$ (Table XX) — roughly two orders of magnitude per basis step. No scalar-versus-relativistic multiplier is quoted here, for the reason given in observation 2; the former "790 scalar → 11,865 spin-ful, 15.0×" used the withdrawn comparator and a retired count. The richer Pauli structure of jj-coupling distributes the Hamiltonian across more distinct Pauli words, so QWC cliques are smaller. This cost is VQE-specific and can be mitigated by symmetry-compressed (SCD) or tensor-factorized measurement strategies.

B. Construction

The angular layer uses Szmytkowski's spinor spherical harmonic reduction [33] in the (κ, m_j) basis, with $\kappa = -(l+1)$ for $j = l + 1/2$ and $\kappa = +l$ for $j = l - 1/2$. One-body matrix elements of $\sigma \cdot \hat{r}$, $\mathbf{J}^2$, $\mathbf{L} \cdot \mathbf{S}$, and σ^2 are closed-form integer / half-integer eigenvalues; rank-1 components of $\mathbf{L}_z$ and σ_z are emitted as symbolic Wigner–Eckart reductions and closed by the spin-orbit consumer (below).

The radial layer reuses the non-relativistic hydrogenic radial integrals $\langle r^k \rangle_{n,l}$ and $\langle 1/r^k \rangle_{n,l}$ as closed-form Bethe–Salpeter rationals in (Z, n, l). The full Dirac-Coulomb radial corrections, which would enter through the Martínez-y-Romero three-term recursion [36] and carry the factor $\gamma := \sqrt{1 - (Z\alpha)^2}$, are *reserved* for a future extension. At Tier 2 they enter only as an unbound symbolic placeholder registered by the spinor certifier (§V E); no T3 matrix element depends on γ.

The two-body ERI block applies Dyall §9.3's jj-coupled Coulomb reduction [31] in its pre-recoupled "separable" form,

$$\langle ab \,|\, 1/r_{12} \,|\, cd \rangle \;=\; \sum_k X_k(a,c)\, X_k(b,d)\, R^k(n_a l_a, n_b l_b, n_c l_c, n_d l_d),$$

where X_k is the full-Gaunt jj-coupled angular coefficient (parity $l_a + l_c + k$ even, j-triangle, and the two 3j bottom-row constraints; see also Paper 22 § 5 "Restatement in the spinor basis"), and R^k is the scalar radial integral evaluated by the same exact-rational `hypergeometric_slater.py` evaluator used by the spinless builder. The explicit 6j symbol from Dyall's *coupled* form factorizes through $X_k(a,c) \cdot X_k(b,d)$ with the $m_{j,a} + m_{j,b} = m_{j,c} + m_{j,d}$ global constraint; the composed pipeline operates in the uncoupled spin-orbital basis, so the separable form is the one consumed.

Spin-orbit is added as a one-body diagonal update [34]:

$$\langle n, \kappa, m_j \,|\, H_{\text{SO}} \,|\, n, \kappa, m_j \rangle \;=\; -\,\frac{Z^4\, \alpha^2\, (\kappa + 1)}{4\, n^3\, l(l + \frac{1}{2})(l+1)},$$

diagonal in (n, κ, m_j) with the Kramers cancellation $H_{\text{SO}} = 0$ at $l = 0$ ($\kappa = -1$ forces the numerator to zero before the formally divergent $\langle 1/r^3 \rangle$ is evaluated). This is purely algebraic: a rational in (Z, n, l) times α^2, no radial quadrature, diagonal. Darwin and mass-velocity corrections are explicitly out of scope at Tier 2; sub-MHz spectroscopic accuracy is not a design goal.

C. Chawla et al. head-to-head

Table XXI compares GeoVac's Tier 2 relativistic Pauli counts to Chawla et al. [37]'s OpenFermion-Dirac / cv2z construction at $Q = 18$. The only fully calibrated Chawla cell available in the main paper is RaH at 18q (12 556 relativistic Pauli strings); per-molecule counts for BeH, MgH, CaH, SrH, BaH at 18q are in the Supplemental Material Tables S1–S3, which are **flagged as deferred** from this comparison (inaccessible in the sprint timeframe).

Native Q gives GeoVac an 8.4–12.6× advantage against RaH-18q. *The matched-Q projection is withdrawn* (2026-08-30): it was built from the retired $O(Q^{2.5})$ composed

TABLE XXI. GeoVac Tier 2 vs. Chawla et al. [37]. Native-Q GeoVac numbers at $n_{\max} = 2$ compared to the single available Chawla cell (RaH-18q). Re-measured 2026-08-30 under the corrected Condon–Shortley angular factor order (retired: 1,413 / 942, native ratios 0.113/0.075). A former "projected at $Q = 18$" column is *withdrawn*: it was extrapolated with the retired $O(Q^{2.5})$ composed scaling law and the 4.2× rel/scalar multiplier that §V A withdraws, so no corrected value exists to replace it. The comparison is at native Q, which differs between the two constructions — see the caveats below.

Molecule	Q (GeoVac)	GeoVac N_{Pauli}	Chawla RaH-18q	Native ratio
LiH	30	1 501	12 556	0.120×
BeH	30	1 501	12 556	0.120×
CaH	20	998	12 556	0.079×

scaling law and the 4.2× rel/scalar multiplier, and that multiplier is the ratio this paper withdraws in §V A as having compared a quantity with itself. With no valid multiplier there is nothing to rebuild the projection from, so only the native-Q ratios are reported; the former claim was ∼ 17–32×. The advantage that remains is structural, not accidental: the composed + frozen-core architecture eliminates the large cv2z primitive basis that dominates Chawla et al.'s 18q Pauli tensor.

Honest caveats on the Chawla et al. comparison.

1. The RaH-vs-LiH / BeH / CaH comparison is not species-matched. The four cells missing from Chawla et al.'s main paper (BeH-18q, MgH-18q, CaH-18q, SrH-18q, BaH-18q) would make this a genuine head-to-head.

2. GeoVac composed accuracy is 5–26 % R_{eq} error at the matched molecules (Paper 17); Chawla et al.'s DC-UCCSD is accuracy-targeted (0.47% vs CASCI for CaH). The comparison is *resource estimation at fixed single-point geometry*, not a spectroscopic benchmark.

3. GeoVac uses Breit–Pauli spin-orbit at leading $Z\alpha^2$; full Dirac-Coulomb $\gamma = \sqrt{1-(Z\alpha)^2}$ radial corrections are reserved. For CaH ($Z = 20$) the relative γ correction to the fine-structure splitting is ∼ 1 %.

A fine-structure sanity check on He, Li, and Be 2p splittings (benchmark code in `benchmarks/fine_structure_check.py`) reproduces all three to correct sign and correct order of magnitude ($-66\%, +211\%, -78\%$ relative error against NIST), consistent with the first-order $Z\alpha^2$ approximation plus simple Z_{eff} screening. Absolute accuracy is not a Tier 2 design target; the 20–50 % target suggested in the Tier 2 Explorer is not met for Li. Reaching sub-10 % accuracy would require multi-electron configuration interaction with Breit-Pauli spin-spin and spin-other-orbit terms, which is Tier 3+ work.

D. Angular sparsity at the spinor level

The potential-independent angular sparsity theorem of Paper 22 extends verbatim to the jj-coupled spinor basis, with the ERI density replaced by the spinor analog $d_{\text{spinor}}(l_{\max})$ of Paper 22's scalar density $d(l_{\max})$. See Paper 22 § 5 "Restatement in the spinor basis" for the full table in both conventions. Briefly: $d_{\text{spinor}} \le d_{\text{scalar}}$ at every $l_{\max}$, with ratio 0.25 at $l_{\max} = 0$ (pure spin dilution) climbing to 0.92 at $l_{\max} = 5$; the sparsity *exponent* is preserved, the *prefactor* is inflated.

At matched $l_{\max}$ the spinor Pauli tensor is ∼ 16× denser than the scalar one (from the $(2Q/Q)^4 = 16$ four-tuple count times $d_{\text{spinor}}/d_{\text{scalar}} \approx 0.86$ at $l_{\max} = 3$). At matched Q (the regime of Table XX) no spinor multiplier is quoted: the comparison that produced the former "11.33× at $n_{\max} = 3$" is withdrawn, since its scalar denominator was not a scalar count. The distinction between matched-$l_{\max}$ and matched-Q accounting matters when comparing to scalar published baselines; all GeoVac numbers in this section are matched-Q.

E. Paper 18 taxonomic classification

Every coefficient in the Tier 2 Hamiltonian lies in the *spinor-intrinsic* ring

$$R_{\text{sp}} \;:=\; \mathbb{Q}(\alpha^2)[\gamma]\,/\,(\gamma^2 + (Z\alpha)^2 - 1),$$

per the new Paper 18 §IV subtier (forthcoming, drop-in in `docs/paper18_spinor_subtier_proposal.tex`). A `verify_spinor_pi_free` certifier (in `geovac/spinor_certificate.py`, Track T5) walks the sympy expression tree of each matrix element and rejects any appearance of π, ζ, log, E_1, or unregistered free symbol. The symbolic spin-orbit block (Track T2) passes the certifier cleanly for all $n_{\max}$ up to 4; manual contamination (e.g., injecting $\pi \cdot \alpha^2/96$ into a coefficient) is rejected with a clear diagnostic.

The Tier 2 transcendental content is therefore entirely algebraic in R_{sp}: no new calibration π, no odd zeta, no embedding constant beyond the existing scalar seeds. The first-order-operator × spinor-bundle cell of Paper 18's operator-order × bundle grid is now populated; the 2nd-order-operator × spinor-bundle cell is flagged as an open slot for future QED-on-S^3 work.

F. Scope and Tier 3 outlook

The spin-ful composed encoding described in this section is a *capability* addition, not a new fundamental projection. It extends the scalar composed pipeline to molecules where spin-orbit coupling matters structurally (first-row: modest; third-row like CaH: ∼ 1% of valence energy at $Z = 20$; fifth-row and heavier: dominant).

Three natural extensions are Tier 3+ work:

1. Frozen-core [Kr] infrastructure for SrH, BaH, RaH — would enable direct point-by-point Chawla et al. comparison at native $Q = 18$ and close the four deferred cells in Table XXI.

2. Martínez-y-Romero Dirac-Coulomb radial recursions to bind γ — would lift Table XX from Breit-Pauli to full-Dirac-Coulomb accuracy. $R_{\rm sp}$ and the certifier already accommodate γ with no ring change required.

3. Darwin and mass-velocity terms plus multi-electron Breit-Pauli spin-spin / spin-other-orbit — would lift the OoM-correct fine-structure splittings of §V C to few-percent accuracy. A production spectroscopic tool; a distinct goal from resource estimation.

No claim is made here that the Tier 2 construction extends to heavy atoms without these Tier 3+ additions; the three extensions listed are the Tier 3 roadmap.

a. Tier 3 fine-structure verification (Tracks T7, T8). The full α^4 one-body fine-structure ladder has been verified exactly. The three terms—spin-orbit (Tier 2, Track T2), Darwin, and mass-velocity (Tier 3, Track T8)—combine to the Dirac formula

$$E_{\rm SO}+E_{\rm D}+E_{\rm MV} \;=\; -\frac{Z^4\alpha^2}{2n^4}\left[\frac{n}{j+\frac{1}{2}}-\frac{3}{4}\right] \quad \text{(Hartree units)}, \tag{18}$$

verified as an exact sympy symbolic equality for all 16 hydrogenic states (n, l, j) through $n = 4$ and all Z tested (including symbolic Z). All six Dirac accidental degeneracies through $n = 4$ are confirmed ($E_{\rm FS}$ depends on (n, j) only, not on l separately). This provides a correctness certificate for the single-particle component of the spin-ful composed pipeline.

Relativistic radial corrections through $\gamma = \sqrt{\kappa^2-(Z\alpha)^2}$ are implemented for $n_r = 0$ states (exact Pochhammer ratios, Track T7); full $n_r \geq 1$ requires the Kramers–Pasternak three-term recursion (deferred).

Fine-structure accuracy for multi-electron He/Li/Be remains at sign-correct and order-of-magnitude level (66–211% relative error in the 2p-doublet splitting); improvement requires multi-electron spin-spin and spin-other-orbit operators (Direction 3, deferred). Darwin + mass-velocity do not improve the 2p-doublet splitting because both j-branches share $l = 1$: Darwin vanishes for $l \geq 1$, and mass-velocity is identical for fixed l.

b. Breit-Pauli SS/SOO fine structure (Track BF). Sprint 3 (Track BF-A+B+C+D, 2026) adds the two-body Breit-Pauli retarded Slater integrals to the framework via a new production module `geovac/breit_integrals.py` (Mellin-regularized radial integrator, 26 tests passing). For the He $2\,^3P$ multiplet with configuration $(1s)(2p)$ the leading Breit-Pauli fine structure takes the form

$$E(^3P_J) \;=\; \tfrac{\zeta_{2p}}{2}\,X(J) \;+\; A_{SS}\,f_{SS}(J) \;+\; A_{SOO}\,f_{SOO}(J), \tag{19}$$

with $X(J) = J(J+1) - L(L+1) - S(S+1)$ and the angular J-coefficients $f_{SS}(J) = (-2, +1, -\frac{1}{5})$, $f_{SOO}(J) = (+2, +1, -1)$ for $J = 0, 1, 2$. The J-patterns are derived symbolically (Track DD, 2026) from the scalar rank-k tensor Racah identity $(-1)^{L+S+J}\,6j\{L\,S\,J;\,S\,L\,k\}$ at $L = S = 1$, with $k = 2$ for SS and $k = 1$ for SOO (normalized so $f(J = 1) = 1$); the derivation and sympy-exact verification are recorded in `tests/test_breit_integrals.py` (`test_drake_f_SS_pattern_from_6j`, `test_drake_f_SOO_pattern_from_6j`). The choice of spin tensor is also structurally fixed by the same 9j algebra: $\langle S{=}1\,\|[s_1\otimes s_2]^{(2)}\|\,S{=}1\rangle = \sqrt{5}/2$ is nonzero (SS uses the coupled rank-2 spin tensor), while $\langle S{=}1\,\|[s_1\otimes s_2]^{(1)}\|\,S{=}1\rangle = 0$ is why SOO must use the sum-of-single-electron tensor $\vec{s}_1 + 2\vec{s}_2$ (Bethe-Salpeter §38.15) rather than the coupled rank-1 form. The radial amplitudes are

$$\begin{aligned} A_{SS} &= \alpha^2\left(\tfrac{3}{50}\,M^2_{\rm dir} - \tfrac{2}{5}\,M^2_{\rm exch}\right), \\ A_{SOO} &= \alpha^2\left(\tfrac{3}{2}\,M^1_{\rm dir} - M^1_{\rm exch}\right), \end{aligned} \tag{20}$$

where $M^k_{\rm dir}$ and $M^k_{\rm exch}$ are the direct and exchange Breit-Pauli retarded Slater integrals $\iint R_{ab}(r_1)R_{cd}(r_2)\,r_<^k/r_>^{k+3}\,r_1^2r_2^2\,dr_1dr_2$ in closed form from the Mellin-regularized evaluator. The combining coefficients $(\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1)$ are pure rationals (Paper 18 intrinsic tier); the radial integrals are rational+embedding-log (Paper 18 embedding tier). The rational direct/exchange mixing ratios were identified by systematic enumeration over small rationals (Track BF-D) and confirmed numerically to reproduce NIST He $2\,^3P$ to sub-percent accuracy. Sprint 5 Track DV (2026) fully characterized the bipolar harmonic expansion of $Y^{(K)}(\hat{r}_{12})/r_{12}^{K+1}$ (Varshalovich §5.17; Brink–Satchler) for the He $(1s)(2p)$ configuration: only a single bipolar channel $(k_1 = 0, k_2 = 2)$ is angular-allowed for the direct path, and only $(k_1 = 1, k_2 = 1)$ is allowed for the exchange path (no higher (k_1, k_2) channels contribute at any $L \geq K$). The piecewise decomposition of the bipolar kernel into Region I ($r_1 < r_2$) and Region II ($r_1 > r_2$) contributions was derived symbolically (using the production module's `_t_kernel_region_I`), and the identity

$$\text{Bipolar}(0, 2, K{=}2)_{\rm dir} = M^0_{\rm dir,I} + M^2_{\rm dir,II}$$

was established, in contrast to Drake's $M^2_{\rm dir} = M^2_{\rm dir,I} + M^2_{\rm dir,II}$. This structural mismatch between the bipolar basis and Drake's M^K basis shows that the rational coefficients $(\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1)$ are a convention-dependent combining identity (specific to how Drake's M^K integrates the piecewise kernel) rather than a direct Racah-algebra output; a fully first-principles derivation remains open. (Sprint 5 Track DV honest-partial; details in `debug/dv_drake_bipolar_memo.md`.)

On the He $2\,^3P$ multiplet with $Z_{\rm nuc} = 2, Z_{\rm eff} = 1$ and

CODATA α, Eqs. (19)–(20) reproduce the NIST splittings to sub-percent accuracy:

Splitting	GeoVac (MHz)	NIST (MHz)	Rel. err.
$E(P_0) - E(P_1)$	+29,612.91	+29,616.95	−0.014%
$E(P_1) - E(P_2)$	+2,231.16	+2,291.18	−2.62%
$E(P_0) - E(P_2)$	+31,844.07	+31,908.13	−0.20%

The 0.20% span error (a 330× improvement over the 66% SO-only T8 baseline, same sign as NIST) meets the Sprint 3 BF-D target of < 20% accuracy. The 2.6% residual on the smaller (P_1, P_2) splitting is at the limit of the four-coefficient rational approximation; further reduction requires finite-mass recoil or higher-order QED corrections (treated in Drake's helium fine-structure reviews; not in scope here).

Extensions to Li $2\,^2P$ and Be $2s2p\,^3P$ (Sprint 3 BF-E) were initially reported as honest negatives at +118% (Li) and 88% (Be span); Sprint 5 Track CP closed both to < 20% via a convention refinement rather than new physics. The standard Breit-Pauli single-particle spin-orbit formula for a valence electron above a closed core is

$$\zeta_{\text{valence}} = \frac{\alpha^2}{2}\, Z_{\text{val}}\, \langle 1/r^3 \rangle_{\text{valence}}, \tag{21}$$

with Z_{val} the *asymptotic* nuclear charge (nuclear charge minus the count of shielding core and same-shell electrons) and $\langle 1/r^3 \rangle = Z_{\text{eff}}^3/[n^3 l(l+\frac{1}{2})(l+1)]$ on a hydrogenic orbital with Slater exponent Z_{eff}. For Li $2p$ above [He]: $Z_{\text{val}} = 3 - 2 = 1$, $Z_{\text{eff}} = 1$ (full-shield hydrogenic), giving $\zeta_{2p} = \alpha^2/48$ and doublet splitting $(3/2)\zeta = 10{,}949$ MHz vs NIST 10,055 MHz, relative error +8.89%. For Be $2s2p\,^3P$: $Z_{\text{val}} = 4 - 3 = 1$ (shielding from [He] plus the $2s$), with Slater's rules $Z_{\text{eff}}(2p) = 4 - 2 \cdot 0.85 - 0.35 = 1.95$; the $E(P_0) - E(P_2)$ span closes to +2.76% of NIST, and all three individual splittings are below 20% (+18.9%, +6.3%, +2.8%). The Sprint 3 BR-C convention used Z_{nuc} (not Z_{val}) in the prefactor; this happens to coincide with the standard form for He (where $Z_{\text{nuc}} = 2 = 2Z_{\text{val}}$ combined with the explicit $(\zeta/2)$ factor in $E_{SO}(J)$), but overcounts by a factor of Z_{nuc} for heavier atoms. The suggested 2-configuration mixing $2s2p \leftrightarrow 2p^2$ is *forbidden by parity*: $2s2p$ has odd parity $(-1)^{0+1} = -1$, while $2p^2$ has even parity $(-1)^{1+1} = +1$, and the parity-even $1/r_{12}$ operator gives zero coupling. This is verified symbolically in `tests/test_breit_integrals.py` (`test_2config_2s2p_2p2_parity_forbidden`). Core polarization (a Migdalek–Bylicki-type model potential, e.g. Phys. Rev. A **24**, 649 (1981), evaluated here with $\alpha_d(\text{Li}^+) = 0.192$ and $r_c = 0.55$ a.u.) is a small correction for Li that *worsens* agreement (from +8.9% to +25%) because it is attractive and increases ζ_{2p}; the Sprint 3 hypothesis that CP was needed for Li was incorrect. The $\sim 8\%$ Li residual is consistent with higher-order $(Z\alpha)^4$ relativistic corrections and QED Lamb shift at the 1% level. The BF-D result is the first sub-percent Breit-Pauli fine-structure benchmark in the framework, and Sprint 5 CP extends the coverage to Li $2\,^2P$ (8.9%) and Be $2s2p\,^3P$ (2.8% span, < 20% on all components) using the same production `geovac/breit_integrals.py` module with no modifications; the extension is a convention cleanup plus Slater's 1930 screening rules.

c. He $2\,^3P$ as the first internal multi-focal precision catalogue entry. The post-MH precision-catalogue arc (Sprint MH 2026-05-08, Track 1 muonium / Ps HFS, Track Mu-Lamb, Track D HFS) tested the multi-focal architecture across the cross-register mass-hierarchy axis (electron + leptonic / I=1/2 / I=1 nuclear registers at vastly different focal lengths) and across the equal-mass limit (positronium). The He $2\,^3P$ Drake-pattern construction documented above is the companion test along the orthogonal *internal multi-focal* axis: two electrons on the same nucleus at structurally distinct effective Z (1s at $Z_{\text{nuc}} = 2$, 2p at full-shield $Z_{\text{eff}} = 1$), coupled via two-body Breit-Pauli SS+SOO operators with bipolar harmonic structure. The Sprint precision-catalogue 2026-05-08 entry adds He $2\,^3P_0$-$2\,^3P_1$ (−0.014%) and the $2\,^3P_0$-$2\,^3P_2$ span (−0.20%) to Paper 34 § V machine-precision, and the small-interval $2\,^3P_1$-$2\,^3P_2$ (−2.62%, partial-cancellation amplification of an absolute −60 MHz residual) to Paper 34 § V.B off-precision (error code A, leading-order Breit-Pauli sitting under the LS-8a α^3/π budget on the absolute scale). This brings the catalogue's coverage matrix to seven precision systems at sub-percent on framework-native parts, spanning mass hierarchy (H, μH, Mu, Ps), nuclear spin (I=1/2, I=1), and now multi-focal kind (cross-register + internal). See `debug/precision_catalogue_he_2_3p.py` and `debug/precision_catalogue_he_2_3p_memo.md` for the sprint provenance.

VI. DISCUSSION

A. Origin of structural sparsity

The Pauli count advantage of the GeoVac lattice is not an artifact of small system sizes or of comparing against minimal Gaussian bases. It arises from a fundamental structural difference in the two-electron integral tensor.

In a Gaussian basis, the molecular orbital (MO) ERIs $\langle pq|rs\rangle$ are obtained by a four-index transformation of the atomic orbital (AO) integrals. Since Gaussian product functions have no angular quantum numbers in the MO basis, there are no selection rules to zero out tensor entries. The MO ERI tensor is generically dense, with $O(M^4)$ nonzero entries.

In the GeoVac lattice, the spin-orbitals carry explicit angular momentum labels (l, m). The two-electron Coulomb operator, expanded in Legendre polynomials, couples orbitals through Gaunt integrals that enforce

$$|l_a - l_c| \le k \le l_a + l_c, \quad l_a + k + l_c = \text{even}, \tag{22}$$

plus the magnetic selection rule $m_a - m_c = m_d - m_b$. These constraints eliminate the vast majority of the M^4 possible ERI entries. The surviving entries grow as $O(M^2)$—a quadratic reduction in the dominant cost term of Eq. (1).

The d-orbital results of Sec. IV L show that the *pair-diagonal* ERI density is lower at higher angular momentum (d-d density 4.0% vs. 8.9% for s+p), because the pair-diagonal rule drops more m-swap integrals as the magnetic grid grows (5×5 for $l = 2$ vs. 3×3 for $l = 1$). This is a property of the *retired pair-diagonal rule*, not the exact selection rule: under the exact global-M_L Coulomb rule the d-block is *denser*, not sparser (Sec. III D). Gaussian bases remain near-dense for transition-metal d-orbitals regardless of rule; the apparent "advantage that grows with angular complexity" for the GeoVac lattice is therefore an artifact of the pair-diagonal approximation, not a genuine structural saving for strongly correlated d-electron systems.

1. Analytical Pauli count for the s+p block

The 279 non-identity Pauli terms per s+p block at $n_{\max} = 2$ ($M = 5$ spatial orbitals, $Q = 10$ qubits) decompose exactly as $55 + 224$: a direct sector and a non-direct sector. (Corrected 2026-08-29. The retired pair-diagonal rule gave $111 = 55 + 56$ from 65 nonzero ERIs; the direct half is unchanged, the non-direct half is four times larger.)

The 55 *direct* terms equal $Q(Q+1)/2$ and are *provably* convention-independent—not merely unaffected by this particular correction. All $M^2 = 25$ Coulomb integrals are nonzero for a reason that bypasses the angular machinery entirely:

$$\langle ab|ab\rangle = \iint \frac{|\phi_a(\mathbf{r}_1)|^2\,|\phi_b(\mathbf{r}_2)|^2}{r_{12}}\,\mathrm{d}\mathbf{r}_1\,\mathrm{d}\mathbf{r}_2 \;>\; 0 \tag{23}$$

strictly, for any orbitals and any basis, the integrand being a product of two non-negative densities and a strictly positive kernel. No sign convention, ordering choice or m-selection rule enters. (The $k = 0$ monopole coefficient $c^0(l, m; l, m) = 1$ is consistent with this but does not establish it on its own: the Slater sum $\sum_k R^k c^k c^k$ carries both signs at $k > 0$, so the monopole term alone does not exclude a cancellation. Positivity does.)

Under Jordan–Wigner encoding the number-operator products $n_p n_q$ map to Z and ZZ strings, giving $\binom{Q}{1} + \binom{Q}{2} = 10 + 45 = 55$ terms. The $\binom{Q}{2} = 45$ two-body (ZZ) coefficients inherit the positivity argument directly: the opposite-spin coefficient is J_{ab}, strictly positive by Eq. (23), and the same-spin coefficient is the antisymmetrized combination

$$J_{ab} - K_{ab} = \tfrac{1}{2}\iint \frac{\left|\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)\right|^2}{r_{12}}\,\mathrm{d}\mathbf{r}_1\,\mathrm{d}\mathbf{r}_2 \;>\; 0 \tag{24}$$

for linearly independent orbitals—again a positive kernel against a non-negative integrand, with no convention entering. So all 45 are convention-independent by proof. The remaining 10 single-Z coefficients carry the one-body h_{pp} alongside J/K sums and admit no such argument; they are nonzero as *measured*, not as proved. These 55 are exactly the all-Z strings of the assembled operator; every one of the remaining 224 carries at least one X or Y.

The 224 non-direct terms come from the 82 non-direct ERIs (107 total minus 25 direct). The nonzero quartet count follows from the exact selection rule rather than from a pair-diagonal count. For each multipole k, let P_k be the set of orbital pairs (a, c) with $c^k(a, c) \neq 0$: the s+p block gives $|P_0| = 7$, $|P_1| = 12$, $|P_2| = 9$. A quartet $\langle ab|cd\rangle$ is nonzero when some k admits *both* pairs *and* the two pairs share a multipole index, i.e. when $m_a + m_b = m_c + m_d$. Taking the union over k of the M_L-constrained products $P_k \times P_k$ gives

$$\left|\bigcup_k \big\{(a, b, c, d) : (a, c), (d, b) \in P_k,\ m_a{+}m_b = m_c{+}m_d\big\}\right| = 107, \tag{25}$$

which is a derivation of the count, not a check against it: the right-hand side is built only from the Gaunt selection rules and the M_L constraint, with no appeal to the assembled tensor. It splits as 59+48 by $|\Delta l|$, and the same construction gives the pure d-shell count as a sum of squared m-multiplicities, $1{+}4{+}9{+}16{+}25{+}16{+}9{+}4{+}1 = 85$. The naive product-of-squares $\sum_k |P_k|^2 = 49 + 144 + 81 = 274$ over-counts on two grounds: it ignores the M_L coupling between the two pairs, and it double-counts quartets admitted by more than one k. The retired $7^2 + 4^2 = 65$ was the corresponding pair-diagonal count.

Because the Gaunt triangle rule requires $l_a + l_c + k$ and $l_b + l_d + k$ both even, the two pairs of an s+p block always carry the same $|\Delta l|$: of the 107 quartets, 59 have $|\Delta l_{ac}| = |\Delta l_{bd}| = 0$ (the $k = 0$ and $k = 2$ channels) and 48 have $|\Delta l_{ac}| = |\Delta l_{bd}| = 1$ (the $k = 1$ dipole channel).

A pure l-shell does *not* collapse to the direct sector. Under the retired rule the pair-diagonal condition forced $a = c$ for every allowed pair, eliminating all exchange terms and giving $Q(Q + 1)/2 = 55$ Pauli terms and Pauli/$Q = 5.5$; that claim is withdrawn. The exact rule constrains only the sum $m_a + m_b = m_c + m_d$, which ordinary exchange integrals $\langle m_1 m_2|m_2 m_1\rangle$ satisfy identically. A pure d-shell ($l = 2$, $M = 5$) therefore carries 85 nonzero ERIs, not 25—60 of them off-diagonal. This is the same reversal reported in Sec. III D and Sec. IV L: the exact rule makes the d-block *denser*, so the reduced coefficient once attributed to transition-metal hydrides has no structural basis.

The universal composed scaling $N_{\text{Pauli}} = n_{\text{sub}} \times 279$ follows from the block-diagonal ERI structure (zero cross-block coupling): $N_{\text{Pauli}}/Q = 279/10 = 27.90$ for any molecule built from s+p sub-blocks at $n_{\max} = 2$.

B. Implications for quantum simulation

The results of Secs. III G–III H show that the three cost metrics—term count, measurement groups, and 1-norm—respond differently to the selection-rule sparsity. Only the term count and the 1-norm benefit; the measurement-group count does not, its exponent (4.01) exceeding the term exponent (3.77) and lying inside the Gaussian $O(Q^{4-5})$ band. For first-order Trotterization, the gate count per step scales as $O(N_{\mathrm{Pauli}} \cdot \bar{w})$ where $\bar{w}$ is the average Pauli weight [12], but the *number of steps* is controlled by the 1-norm (Eq. 8). Because $\lambda \sim Q^{1.77}$ grows far more slowly than $N_{\mathrm{Pauli}} \sim Q^{3.8}$, the total gate count is more favorable than a naive term-count analysis would suggest.

For VQE on near-term hardware, the shot budget scales with the number of QWC measurement groups (Sec. III G). The $O(Q^{4.01})$ group scaling (re-measured 2026-08-30; retired: $O(Q^{3.36})$), combined with the coefficient concentration noted above, implies that the GeoVac lattice basis provides a 1-norm advantage but *not* a measurement-group one: smaller variance per shot (from coefficient concentration), and fewer Trotter steps if Hamiltonian simulation is used as a subroutine [4].

For fault-tolerant algorithms based on qubitization or quantum signal processing, the dominant cost is the number of queries to a block-encoding of H, which scales as $O(\lambda t/\epsilon)$ for simulation time t and error ϵ [12]. The $Q^{1.77}$ 1-norm scaling directly reduces this query count relative to dense-basis encodings with $\lambda \sim Q^{2+}$. The commutator-based analysis (Sec. III H) confirms that the advantage is structural: selection-rule sparsity reduces the fraction of anticommuting Pauli term pairs ($\sim Q^{-0.45}$), not just the coefficient magnitudes, yielding a tighter effective scaling of $r \sim Q^{1.47}$ for first-order Trotterization.

C. Connection to other sparse approaches

Several recent proposals exploit tensor structure to reduce quantum chemistry resource requirements. Double factorization [6] and tensor hypercontraction [7] decompose the ERI tensor to reduce the effective number of terms, achieving $O(M^2)$–$O(M^3)$ scaling in the number of unique rotations. Our approach is complementary: rather than compressing a dense tensor after the fact, the GeoVac lattice produces a natively sparse tensor. The two strategies could in principle be combined—applying low-rank factorization to an already-sparse ERI tensor—though we leave this exploration to future work.

Qubit-ADAPT-VQE [14] and hardware-efficient ansätze [15] also aim to reduce circuit depth by limiting the operator pool. The GeoVac lattice provides a physically motivated basis for such pools: only operators corresponding to nonzero Hamiltonian terms need be included, naturally pruning the search space.

D. Composability with existing reduction techniques

A natural question is whether the structural sparsity reported here composes with existing Pauli reduction techniques, or whether it substitutes for them. The answer is that these approaches operate on orthogonal axes and their benefits are additive.

Qubit tapering [3] exploits $\mathbb{Z}_2$ symmetries to eliminate redundant qubits. The GeoVac lattice Hamiltonian preserves S_z and particle-number symmetry, so the same 2-qubit reduction available to Gaussian encodings applies. Indeed, the published Gaussian baselines of Trenev *et al.* [29] already use this reduction; GeoVac's advantages are reported *without* it, making the comparisons conservative.

1. Hopf-$U(1)$ tapering: per-sub-block qubit reduction unique to the natural-geometry basis

The $\{Z_\alpha, Z_\beta\}$ pair (particle number and S_z) is available to any Jordan–Wigner encoded chemistry Hamiltonian. GeoVac admits a further $\mathbb{Z}_2$ *per orbital sub-block* that Gaussian encodings break structurally: the Hopf-$U(1)$ $m_\ell \to -m_\ell$ reflection identified in Paper 29 [25] §5.3 (Observation 5.1) as the only sub-action of the continuous Hopf-$U(1)$ of Paper 25 [24] that commutes with a real-integer adjacency. In the standard composed builder the ERI tensor is block-diagonal across sub-blocks (`composed_qubit.py:727--745`), so the $m_\ell \to -m_\ell$ reflection acts independently within each sub-block, giving one commuting $\mathbb{Z}_2$ stabiliser P_i per sub-block with at least one $\ell \geq 1$ shell. For a molecule with n_{sb} such sub-blocks (the universal case at $n_{\max} \geq 2$, where every block carries a $2p$ shell), the total qubit reduction is

$$\Delta Q = 2 + n_{\mathrm{sb}}. \tag{26}$$

Applied across the 37-system library covered by the `ecosystem_export` interface, this gives ΔQ ranging from -3 (He, H_2; $n_{\mathrm{sb}} = 1$) to -12 (CO, N_2, F_2; $n_{\mathrm{sb}} = 10$), with total library qubit savings of 254 (mean -6.9 per molecule; the library sum is pinned by `tests/test_z2_tapering.py`) and machine-precision spectrum preservation ($|\Delta E_{\mathrm{GS}}|/|E_{\mathrm{GS}}| \leq 4 \times 10^{-15}$, measured at v2.6.0 for every system small enough to diagonalise the naive Hamiltonian sparsely at $Q_{\mathrm{naive}} \leq 20$: H_2, He, NaH, KH, SrH, BaH; the H_2 and He cases are continuously verified by `tests/test_z2_tapering.py`). The remaining 31/37 systems rely on the structural argument: the basis rotation is orthogonal, the $\{P_i\}$ are mutually commuting Pauli Z-strings with disjoint qubit supports (per-stabiliser commutator residual $< 10^{-13}$ verified numerically per molecule), and projecting onto the $+1$ eigenspace of a Hamiltonian-commuting stabiliser preserves the spectrum exactly on that sector. Unlike the $n_{\mathrm{sb}} = 1$ special case (atom or single-block molecule), where the orthogonal rotation does not merge Pauli strings at scale, the per-sub-block tapering produces a substantial $\sim 12.6\%$ proportional reduction in Pauli term count across the panel (e.g. CO: 1110 $\rightarrow$ 970; BeH_2: 555 $\rightarrow$ 485; H_2O: 777 $\rightarrow$ 679).

In the native (n, ℓ, m) orbital basis the reflection is a permutation, not a Pauli Z-string. An orthogonal rotation to the symmetric / antisymmetric combinations $\phi_\pm = (\phi_{n,\ell,m} \pm \phi_{n,\ell,-m})/\sqrt{2}$ turns it into the JW Z-string $P_i = \prod_{p \in \text{antisym}(i)} Z_{2p} Z_{2p+1}$ per sub-block i, which commutes with the rotated Hamiltonian by construction (cross-block ERIs vanish in the standard composed builder; within each sub-block the Gaunt selection rule $m_1 + m_2 = m_3 + m_4$ is enforced at the ERI tensor level). Together with $\{Z_\alpha, Z_\beta\}$ this yields $2 + n_{\text{sb}}$ commuting stabilisers.

The reduction is library-universal because the symmetry is basis-intrinsic. Gaussian encodings break the $\mathbb{Z}_2$ structurally via the multi-center AO orthogonalisation that mixes m-channels; only an angular-momentum eigenbasis that does not cross-center-mix—such as GeoVac's natural composed build—preserves the symmetry bit-exact, and only such a basis admits the further decomposition into n_{sb} independent per-sub-block reflections. The $\mathbb{Z}_2$'s that survive JW encoding are a direct structural consequence of the framework's geometric origin (the Hopf bundle structure of S^3), not accidental basis-choice symmetries.

The scaling pattern is notable. Most tapering schemes give diminishing returns at scale; for the GeoVac composed encoding specifically, ΔQ scales linearly with sub-block count, which scales linearly with chemical complexity (atoms, bonds, lone pairs). The largest beneficiaries are precisely the production-relevant 100-qubit diatomics (CO, N_2, F_2 at $\Delta Q = -12$), where qubit count is the binding constraint for NISQ and early-FT simulation. The per-sub-block construction degrades gracefully under future extensions: if cross-block ERIs become non-zero (e.g. `cross_block_h1=True`, or balanced-coupled multipole), the per-stabiliser commutator audit catches the offending P_i and drops it from the stabiliser list, recovering the global $n_{\text{sb}} = 1$ result rather than producing an incorrect spectrum. Per-molecule tapering panel documented in `debug/sprint_z2_per_subblock_memo.md` (data file `debug/data/sprint_z2_per_subblock.json`; same-day predecessor at `debug/sprint_z2_tapering_memo.md` covers the $n_{\text{sb}} = 1$ global construction).

Triply-additive blocking with spin adaptation. The Hopf-$U(1)$ $\mathbb{Z}_2$ stabilisers $\{P_i\}_{i=1}^{n_{\text{sb}}}$ commute pairwise with the total spin operator S^2 and with the particle-number-and-S_z stabilisers $\{Z_\alpha, Z_\beta\}$, bit-exactly on LiH, BeH_2, and H_2O (every commutator coefficient is exact zero across 21, 36, and 55 pairs respectively; see `debug/sprint_qpt_hopf_stacking_memo.md`). The Hopf tapering therefore stacks additively with S^2-spin-adaptation techniques such as the recent Quantum Paldus Transform [26], which block-diagonalises a JW Hamiltonian by $u(d) \times SU(2)$ irrep at $O(d^3)$ Toffoli depth. On LiH (4 electrons, 3 active Hopf sub-blocks), the joint $(N, S_z, P_{\text{Hopf}})$ sector is 5.7$\times$ smaller than the particle-number-only sector of standard Bravyi 2017 tapering [3] (4,791 Slater determinants vs 27,405), with further reduction available from full QPT adaptation to S^2 eigenstates. At spectral-triple level the structural reason is that $\{P_i\}$ acts on spatial labels only (the $m_\ell \rightarrow -m_\ell$ reflection within each sub-block), while S^2 acts on spin labels only; the two operators factor cleanly through the spatial–spin tensor decomposition of the GeoVac one-body bundle. The compatibility extends to the spinor-coupled Tier 2 builder (Sec. V) with J^2 replacing S^2: on LiH relativistic at CODATA α, $\|H_{\text{rel}}(\alpha) - H_{\text{rel}}(0)\|_1/\alpha^2 = 3.5417$ bit-identical at $\alpha/2, \alpha, 2\alpha$ (ratio max/min = 1.0000). The natural spinor-side analog of the non-relativistic $m_\ell \rightarrow -m_\ell$ Hopf reflection — the $m_j \rightarrow -m_j$ Z_2 stabiliser on the relativistic Dirac basis — does NOT commute with H_{rel} (residuals 1.7×10^{-2} to 6.8×10^{-2} on LiH/BeH/CaH_{rel} in both the original Dirac basis and after the (sym, antisym) rotation; densification check bit-exact at 0, ruling out implementation bug). The structural reason is the load-bearing finding of this section's tapering story: the non-relativistic m_ℓ Z_2 is *protected* by the integer-ℓ Gaunt parity selection rule, whereas the half-integer-j jj-coupled chemistry has no analog (see Sec. VI D 4). The relativistic-tapering frontier is therefore closed for Z_2 candidates at sprint scale; non-Z_2 machinery (j^2 sector restriction analog of S^2; Kramers projection) remains a separate research direction. Across the 18-molecule hydride library at $n_{\max} = 2$, the per-molecule QPT cost ($\sim M^3$ Toffoli, range 1,000–91,125, total 561,000) is 4–6 orders of magnitude below typical fault-tolerant QPE budgets [7, 32] (10^9–10^{11}), making spin-adaptation effectively free at GeoVac scale.

2. ℓ-parity Z_2: direct Gaunt-parity consequence (strict improvement over Hopf-only)

The Gaunt selection rule for the chemistry ERI requires $(\ell_a + \ell_c + k)$ even AND $(\ell_b + \ell_d + k)$ even on every nonzero $(ab|cd)$, which forces $\ell_a + \ell_b + \ell_c + \ell_d$ to be even on every nonzero ERI element. Equivalently, the operator $P_\ell^{(i)} = (-1)^{N_{\ell\text{-odd}}^{(i)}}$, counting electrons in ℓ-odd orbitals of sub-block i, commutes with the two-body Coulomb sector exactly. Within-block h_1 in the composed builder is ℓ-diagonal so $[P_\ell^{(i)}, h_1] = 0$ trivially.

This is an additional $\mathbb{Z}_2$ stabiliser per sub-block, GF(2)-independent of the Hopf m_ℓ stabiliser of Sec. VI D 1 (an $\ell = 1$, $m_\ell = 0$ orbital carries Hopf parity +1 but ℓ-parity −1). No basis rotation required: orbitals already carry definite ℓ, so $P_\ell^{(i)}$ is a JW Z-string directly. Production module `geovac/extended_tapering.py` ships this as `use_ell_parity=True` (default).

Verification panel (composed builder, $n_{\max} = 2$):

System	$Q_{\rm naive}$	$Q_{\rm Hopf}$	$Q_{\rm Hopf} + \ell$	ΔQ_ℓ	Pauli ratio
LiH	30	25	**22**	+3	0.97
HF	60	52	**46**	+6	0.97
BeH_2	50	43	**38**	+5	0.97
H_2O	70	61	**54**	+7	0.97
NH_3	80	70	**62**	+8	0.97
CH_4	90	79	**70**	+9	0.97

$\Delta Q_\ell = n_{\rm sb}$ (one stabiliser per sub-block with ℓ-odd content) is matched bit-exactly across the panel. The Pauli-count ratio is uniformly ~ 0.97 (a $\sim 3\%$ reduction beyond Hopf-only), making this a *strict improvement* over Hopf-only on both axes. Ground-state spectrum preservation is verified on H_2: the minimum over tapered sector choices matches the naive JW ground energy to $< 10^{-10}$ (test `tests/test_extended_tapering.py`, `test_extended_hopf_ell_spectrum_preserved_h2`).

3. *Hidden per-sub-block particle-conservation Z_2: meta-investigation via symmetry-adapted basis*

The combined $\mathbb{Z}_2^3$ irrep decomposition under $(p_{\rm Hopf}, p_\ell, p_{\rm swap})$ exposes additional hidden $\mathbb{Z}_2$ symmetries that the piecewise stabiliser search of Secs. VI D 1–VI D 2 would not surface. Concretely, in the composed builder the ERI is block-diagonal across sub-blocks (no cross-block ERI by default, `composed_qubit.py:727--745`), so the per-sub-block electron count N_b is conserved *individually*, and

$$P_N^{(b)} = (-1)^{N_b} \tag{27}$$

is a $\mathbb{Z}_2$ stabiliser commuting with H, independent in GF(2) of the Hopf m_ℓ stabiliser (which only Z-strings the ℓ-odd antisymmetric orbitals) and of the ℓ-parity stabiliser (which only Z-strings the ℓ-odd orbitals). $P_N^{(b)}$ Z-strings *all* orbitals in block b, a strictly larger support. Production module `geovac/symmetry_adapted_basis.py` ships this with the `extended_plus_hidden_tapered_from_spec` entry point; wired into the production API at `ecosystem_export.hamiltonian(tapered='full')`.

Across the same panel (extending Sec. VI D 2):

System	Q Hopf+ℓ	Q full hidden	ΔQ
LiH	22	**20**	+2
HF	46	**41**	+5
BeH_2	38	**37**	+1
H_2O	54	**53**	+1
NH_3	62	**55**	+7
CH_4	70	**62**	+8
Total	292	**268**	+24

The audit gate verifies GF(2)-independence of each candidate stabiliser; 24 of 132 panel candidates pass and contribute. Ground-sector agreement between the extended and extended+hidden tapered forms is verified to $< 10^{-10}$ Ha on LiH max_n=1 via full diagonalisation, alongside block-diagonality and stabiliser-audit checks (test `tests/test_symmetry_adapted_basis.py`); the naive-vs-tapered spectrum comparison is carried by the H_2 ℓ-parity test above. Combined with Hopf $+\ell$-parity, the production `tapered='full'` pipeline saves $30 \to 20$ on LiH (33% reduction) and $90 \to 62$ on CH_4 (31%) over naive JW — a multiplicative win across the chemistry-encoding-relevant $\sim Q = 30$–100 qubit regime.

4. *Structural reason: Gaunt parity protects integer-ℓ parity Z_2's but not their half-integer-j analogs*

The Hopf m_ℓ Z_2 (Sec. VI D 1) and the ℓ-parity Z_2 (Sec. VI D 2) both commute with H for the *same* underlying structural reason: under all-orbital sign-flip of the parity quantum number in a 4-orbital ERI element, the angular factor accumulates a sign $(-1)^{\sum_i p_i}$ from the four $3j$-symbol parity factors. For integer ℓ with sign flip $m_\ell \to -m_\ell$ (Hopf) the multipole Gaunt rule $(\ell_a + \ell_c + k)$ even AND $(\ell_b + \ell_d + k)$ even forces $\ell_a + \ell_b + \ell_c + \ell_d$ even. For integer ℓ with the ℓ-parity sign flip directly, the same rule forces the analog sum even. Both Z_2's are *protected* by the Gaunt parity selection rule on the multipole-expanded ERI: the four-orbital sign factor is *forced* to be +1, ensuring the ERI tensor is invariant under the reflection.

The half-integer-j jj-coupled analog has no such protection. Under all-orbital $m_j \to -m_j$ flip the four 3j-symbol sign factors give $(-1)^{j_a+j_b+j_c+j_d}$, and the jj-coupled selection rules do not force this sum to be even: a quadruple $(j_a, j_b, j_c, j_d) = (3/2, 1/2, 1/2, 1/2)$ has sum 3 (odd), factor -1, and such ERI elements are present at non-zero amplitude in $H_{\rm rel}$. This is the load-bearing structural reason the relativistic m_j Z_2 candidates fail (both direct-basis and rotated-basis; see the 1.7–6.8×10^{-2} residuals reported above).

The general principle is: for any parity-style $\mathbb{Z}_2$ tapering candidate in a chemistry construction, the candidate commutes with the two-body Coulomb sector if and only if the four angular-momentum-parity factors are *forced* to be +1 by the ERI's angular selection rules. Integer-ℓ Gaunt parity provides this protection cleanly (yielding the Hopf + ℓ-parity Z_2's); half-integer-j jj-coupling does not. This is the underlying reason GeoVac's chemistry-side computational story (O($Q^{2.5}$) Pauli scaling, $\sim 30\%$ qubit reduction at production via four-axis tapering, Pauli-count reduction) is the non-relativistic composed builder rather than its relativistic Tier 2 sibling.

5. *MPO bond rank: structural inheritance from the multipole expansion*

The operator Schmidt rank χ_k at qubit cut k of a chemistry Hamiltonian under Jordan–Wigner ordering equals the MPO bond dimension after maximally compact decomposition. For chemistry Hamiltonians this can be computed directly from the Pauli expansion without running DMRG, by SVD of the coefficient matrix $M[P_\mathrm{L}, P_\mathrm{R}]$ that decomposes each Pauli string into left and right parts at the cut.

Bond rank decomposition of the GeoVac composed Hamiltonian (Thm. 3.2.A.unified). Let $H = h_1 + V_{ee}$ be a GeoVac composed Hamiltonian on N spin-orbitals. At every Jordan–Wigner cut $k \in \{1, \ldots, N-1\}$, the operator Schmidt rank χ_k decomposes into five components.

(A) One-body exact closed form.

$$\chi_k^{h_1} = 2\,\mathrm{rank}\big(h_{p\le k,\,q>k}\big) + \mathbf{1}_{LL} + \mathbf{1}_{RR}, \tag{28}$$

where $h_{p\le k,\,q>k}$ is the $k \times (N-k)$ cross-cut sub-matrix of h_1 in spin-orbital indexing and $\mathbf{1}_{LL}, \mathbf{1}_{RR}$ indicate the existence of nonzero pure-left and pure-right one-body content. The factor of 2 is the JW $XX + YY$ structure of the cross-cut hopping.

(B) Two-piece subadditivity.

$$\chi_k^H \le \chi_k^{h_1} + \chi_k^{V_{ee}}, \tag{29}$$

with slack absorbed by additive baseline mode sharing between h_1 and V_{ee}.

(C) Per-multipole subadditivity.

$$\chi_k^{V_{ee}} \le \sum_L \chi_k^{V_{ee}^L}, \tag{30}$$

where V_{ee}^L is the L-multipole component of V_{ee} under the Slater–Gaunt decomposition (Sec. II). Slack absorbs cross-L operator overlap when a fermionic pair $a_p^\dagger a_q$ carries weight in multiple Gaunt-allowed L.

(D) F3 inheritance. The per-L rank $\chi_k^{V_{ee}^L}$ inherits the graded singular-value spectrum of the radial-integral matrix $R^L(\rho_\alpha; \rho_\beta)$ that controls the double-factorization rank reported in Sec. I (the same hydrogenic-basis Laguerre-recurrence ill-conditioning).

(E) Boundary saturation. At every sub-block boundary the composed builder has zero cross-block ERI content (F4 of the DF–multipole equivalence), all cross-cut content vanishes, and $\chi_k^H = 2$ (additive baseline of $H_\text{left block} \otimes I + I \otimes H_\text{right block}$).

Universal interior profile. For the standard $\{1s, 2s, 2p_{-1,0,+1}\}$ valence at $n_\mathrm{max} = 2$, the interior bond rank within a sub-block follows the universal core sequence $\chi_k^H = \{16, 16, 9, 9, 9, 6, 3\}$ at local cuts $k = 2, \ldots, 8$, depending only on the orbital basis; the edge cuts are position-dependent (first cut: 4 for the leading sub-block, 5 for sub-blocks with a completed block to their left — one extra additive baseline mode; the final sub-block's last cut saturates to the chain-end value 2). The leading-sub-block sequence at cuts $1, \ldots, 10$ is $\{4, 16, 16, 9, 9, 9, 6, 3, 3, 2\}$. Verified bit-exactly on LiH at 29/29 cuts (recomputed from the live builder and pinned by `tests/test_paper14_mpo_rank.py`); the $\chi_k = 2$ boundary saturation is verified across LiH (3 sub-blocks), BeH_2 (5 sub-blocks), and H_2O (7 sub-blocks) at 12/12 sub-block boundaries, giving 8× interior/boundary ratio. The dominant rank carrier transitions with the cut position: $L = 0$ governs cuts 1–3 (1s–2s cross-cut active), $L = 1$ governs cuts 4–6 ($1\mathrm{s}, 2\mathrm{s} \leftrightarrow 2\mathrm{p}$ active), $L = 2$ governs cuts 7–9 (sequential 2p closure). The profile is determined by the orbital structure and the multipole hierarchy ($L_\mathrm{max} = 2\ell_\mathrm{max}$), with no fit parameters.

Bit-exact verification panel. On LiH, statement (A) holds at 29/29 cuts (eq. (28) exact); (B) and (C) hold at 29/29 with average slack 2.3 and 5.9 respectively, both from shared additive baseline modes between h_1, V_{ee}^L. Driver and data files (permanent): `tests/mpo_rank_support/sprint_s2_v2_unified_panel.py` + `tests/mpo_rank_support/data/sprint_s2_v2_unified_panel.j` the full LiH panel is recomputed from the live builder and the shipped panel pinned by `tests/test_paper14_mpo_rank.py` (sprint chronicle: CHANGELOG v3.54.0). Negative control extended to the balanced-coupled library: across LiH, BeH_2, H_2O at 12/12 sub-block boundaries the chi= 2 floor is broken into a clean two-value quantization $\chi_k^{H,\,\text{balanced}} \in \{9, 16\}$ that decomposes as $\chi_k^{H,\,\text{balanced}} - 2 = N_\text{cross}(k) \times 7$, where $N_\text{cross}(k) \in \{1, 2\}$ counts cross-block multipole couplings straddling the boundary and 7 equals the double-factorisation rank of the $\{1s, 2s, 2p\}$ valence basis (Sec. I). The boundary lift therefore directly inherits the F3 graded structure: the same Laguerre–Gaunt ring that fixes DF rank fixes the per-boundary lift, bit-identically across three molecules and twelve boundaries (Sprint S1 / S2-v2 negative controls; panel data `tests/mpo_rank_support/data/sprint_s2_v2_balanced_library_p` pinned by `tests/test_paper14_mpo_rank.py`; chronicle CHANGELOG v3.54.0).

This bond-rank inheritance is a chemistry-side manifestation of the spectral-truncation GH-convergence structure underlying the GeoVac spectral triple: the L-multipole density operators are the natural carriers of the operator system at the cut, and the MPO bond rank inherits the graded rank of the radial-integral matrix R^L from the same Laguerre–Gaunt ring that supplies double-factorization compression (Sec. I). Properties (A)–(E) above provide a quantitative DMRG-side realisation of the qualitative "$V_{ee} \in \mathcal{A} \otimes \mathcal{A}$" statement of Sec. I.

Pauli grouping (QWC, general commutativity, unitary partitioning [4, 5]) reduces measurement overhead by identifying simultaneously measurable Pauli strings. These algorithms take a set of Pauli strings as input and group them; a smaller input set produces fewer groups. GeoVac provides the smaller input set—$Q^{3.8}$ instead of Q^{4+} terms—so grouping starts from a more favorable position.

Low-rank factorization [6, 7] decomposes the ERI tensor to reduce the number of distinct rotation circuits. The rank of the GeoVac ERI tensor is bounded by the number of allowed Gaunt channels, which grows as $O(l_\mathrm{max}^2)$—already much lower than the $O(M^2)$ rank of a generic dense tensor. Applying low-rank factorization to a natively sparse tensor would start from a lower rank, potentially achieving further compression.

In short, GeoVac's structural sparsity provides a better starting point for every downstream optimization in the standard quantum chemistry compilation pipeline. The $Q^{3.17}$ composed scaling (Sec. IV) is a *floor* that existing reduction techniques can only improve upon.

E. Fault-tolerant resource implications

For fault-tolerant quantum simulation, the total cost is controlled by the product of gate count per Trotter step and the number of steps r. We emphasize two structural advantages of the GeoVac encoding in this regime.

First, the sub-quadratic 1-norm scaling ($\lambda \sim Q^{1.77}$) means that the GeoVac advantage *grows faster* for fault-tolerant algorithms than for VQE. The Trotter step count $r \propto \lambda/\sqrt{\epsilon}$ depends on 1-norm, not term count; the composed 1-norms (Table XI) show that LiH at $Q = 84$ requires $r \approx 4{,}500$ steps at chemical accuracy ($\epsilon = 10^{-3}$), while BeH_2 at $Q = 140$ requires $r \approx 16{,}500$—modest counts for early fault-tolerant devices.

Second, the commutator-based bound (Table VII) provides a further 4–7× step reduction that also grows with system size ($r_{\mathrm{comm}} \sim Q^{1.47}$ vs. $r_{\lambda^2} \sim Q^{1.77}$). This reflects the physical mechanism: selection-rule sparsity ensures that an increasing fraction of Pauli term pairs commute, reducing the effective error per step.

For qubitization [7, 12], the query complexity scales as $O(\lambda t/\epsilon)$ with no quadratic dependence. The $Q^{1.77}$ 1-norm scaling translates directly into a $Q^{0.3–0.5}$ query-count advantage over encodings with $\lambda \sim Q^{2+}$. At $Q = 110$, this corresponds to a $\sim 3\times$ reduction in queries; at $Q = 1000$ (extrapolated), the advantage would grow to $\sim 10\times$.

These advantages compose with the Pauli term count reduction: fewer terms means fewer gates per Trotter step, and lower 1-norm means fewer steps. The product $N_{\mathrm{Pauli}} \times r$ determines the total gate budget, and both factors are simultaneously reduced by the GeoVac encoding.

F. Context: fault-tolerant Gaussian compression methods

The fault-tolerant QPE literature employs block-encoding methods—double factorization (DF) [6], tensor hypercontraction (THC) [7], and symmetry-compressed double factorization (SCDF) [27]—that do not produce Pauli decompositions. These methods target the block-encoding 1-norm λ and Toffoli gate count rather than Pauli term count, and are implemented as sequences of rotations within a linear combination of unitaries (LCU) circuit. Published benchmarks focus on large active spaces: FeMoCo at 108–152 qubits [7], cytochrome P450 at 96–116 qubits [28], and similar production-scale systems, with further reductions reported through improved tensor factorization and active volume compilation [28]. To our knowledge, no published DF, THC, or SCDF λ values exist for molecules at the 10–70 qubit operating scale of the present work (LiH at $Q = 30$, H_2O at $Q = 70$), because these systems are too small for the overhead of block-encoding to be beneficial. The Pauli term advantage reported here ($\sim 50\times$–$320\times$, Sec. IV D; the retired pair-diagonal rule gave $51\times$–$1{,}712\times$) is therefore against raw Jordan–Wigner encoding, the representation used by VQE and other near-term variational algorithms. The 1-norm comparison (Sec. III H) and its scaling ($\lambda \sim Q^{1.77}$) are the metrics relevant for both VQE shot budgets and fault-tolerant qubitization query complexity.

a. Computed 1-norm comparison. To quantify the 1-norm comparison directly, we compute Gaussian raw Jordan–Wigner baselines from published molecular integrals. For LiH STO-3G (6 spatial orbitals, $Q = 12$, FCI $E = -7.881$ Ha), the raw JW Hamiltonian has 907 Pauli terms, $\lambda = 34.3$ Ha, and 273 QWC groups. The composed GeoVac LiH ($Q = 30$, electronic-only with PK partitioned classically) has 838 Pauli terms, $\lambda = 34.0$ Ha, and 64 QWC groups (measured 2026-08-30 under the exact rule; the retired pair-diagonal rule gave 334, 32.6 Ha and 21). At matched raw-JW convention—not matched qubit count (GeoVac uses 2.5× more qubits, $Q = 30$ vs. 12) and not matched accuracy (the composed LiH carries ~5% R_{eq} geometry error and does not bind under qubit FCI)—GeoVac is at *near parity* on Pauli terms (1.08× fewer, 838 vs. 907) with 4.3× fewer QWC groups and essentially identical 1-norm (0.99×). Under the retired rule this read "2.7× fewer Pauli terms and 13× fewer QWC groups"—a decisive-looking win that the correction removes. The 1-norm parity reflects that λ is dominated by one-body terms (core energies, PK), which scale with nuclear charge rather than orbital count. The Pauli and QWC advantage are the VQE-relevant metrics. For the validated He equal-qubit comparison at $Q = 28$, the GeoVac lattice has 7.1× lower 1-norm than Gaussian cc-pVTZ ($\lambda = 74.21$ vs. 530.47 Ha), demonstrating that the 1-norm advantage grows with basis size for single-center systems.

b. BLISS-THC at the production-scale ceiling. For context on the FTQC operating scale, Caesura *et al.* [28] report the current best-of-class BLISS-THC factorization for cytochrome P450 in a (63e, 58o) active space ($Q = 116$ spin-orbitals): 999 memory qubits, $\lambda_{\text{BLISS-THC}} \approx 133$ Ha, and 1.71×10^9 Toffoli gates after circuit improvements; the prior-art THC of Goings *et al.* [8] reported $\lambda_{\mathrm{THC}} = 388.9$ Ha and 7.79×10^9 Toffolis at the same active space. Caesura's ~233× runtime speedup over baseline THC combines a factor of ~25 from Active Volume photonic compilation, a factor of ~8 from BLISS within THC, and ~1.1 from block-encoding circuit improvements; the λ improvement alone is roughly 3×, bringing the BLISS-THC 1-norm to within a factor of 2 of the symmetry-shifted lower bound 69.3 Ha for P450 [28]. These resource estimates are reported for a single P450 active space, the smallest molecule in the Caesura benchmark; the GeoVac multi-center library reaches CO and N_2 at $Q = 100$ with 2,790 Pauli terms under the composed encoding (Sec. IV K), but Pauli term count and 1-norm/Toffoli count are not directly comparable metrics — Pauli count drives VQE measurement overhead and Trotterization step depth, while the BLISS-THC pipeline optimizes the qubitization 1-norm and Toffoli budget through a non-Pauli (block-encoded LCU) circuit family. The two operating points target different algorithmic regimes; the appropriate head-to-head comparison would require running BLISS-THC on a GeoVac-scale molecule (LiH, H_2O), which is below the published benchmark range, or running GeoVac at P450 scale, which is above the current library ceiling.

G. Limitations

Several caveats apply to the present results.

System size. The scaling exponents are extracted from four data points at $Q = 10$–110 qubits. While the ERI density decay ($\approx M^{-0.49}$ under the exact rule) has an analytical foundation in the Gaunt integral selection rules, the precise exponents ($\alpha_{\text{terms}} = 3.77$, $\alpha_\lambda = 1.77$) may shift slightly as larger systems (Li, Be, molecular H_2 at higher $n_{\max}$) are included. All R^2 values exceed 0.997, indicating good power-law behavior over the range tested.

Gaussian baseline. The genuine Gaussian integrals (Sec. III I) now cover He at three basis sizes (STO-3G, cc-pVDZ, cc-pVTZ), providing validated equal-qubit comparisons at $Q = 2, 10, 28$. At the two largest sizes the 1-norm advantage holds—3.8× and 7.1× lower—but the Pauli-term comparison does *not* run one way: at $Q = 10$ GeoVac has 1.8× *more* terms, recovering to 1.5× fewer at $Q = 28$. (Corrected 2026-08-30; the retired pair-diagonal rule gave 1.3× and 8.1× fewer terms and 7.1× lower 1-norm at $Q = 28$.) The 6-31G entry in Table I still uses synthetic integrals with representative sparsity. The molecular Gaussian baselines use the published Pauli counts of Trenev *et al.* [29] (Table 5, Appendix B). The structural conclusion—Gaussian ERIs are near-dense while lattice ERIs are selection-rule-sparse—is robust and well established [17].

1-norm bound. The Trotter step counts in Table VI use the standard λ^2 upper bound (Childs et al. [12], Eq. 48). As shown in Table VII, commutator-based bounds that exploit the detailed Hamiltonian structure yield 7× fewer Trotter steps at $Q = 60$, with scaling exponent $\alpha = 1.47$ versus 1.77 (the 1.69 previously quoted here was the retired λ exponent). The commutator sweep covers $n_{\max} = 2$–4 (three data points); the $n_{\max} = 5$ system ($N = 2{,}434{,}441$ terms under the exact rule; retired: 227,338) requires an $N \times N$ anticommutativity matrix of order 10^{12} entries (~ 6 TB as bytes) and was not computed. The ~ 52 GB figure previously quoted here was $227{,}338^2$, i.e. computed from the retired term count. The scaling exponent may shift slightly with additional data points.

Orbital ordering. Jordan–Wigner Pauli weights depend on the spin-orbital ordering. We use the natural lattice ordering (n, l, m, σ) sorted lexicographically. Optimized orderings (e.g., interleaving by angular momentum) could further reduce the average Pauli weight without changing the term count.

Beyond two-electron integrals. For systems with strong relativistic effects or explicit correlation factors, additional integral classes arise that are not captured by the present analysis. The selection rule argument applies specifically to the nonrelativistic Coulomb ERI tensor.

VII. CONCLUSION

We have demonstrated that qubit Hamiltonians constructed from spectral graph theory lattices produce structurally sparse Pauli decompositions across three operationally relevant cost metrics:

- **Pauli term count:** $O(Q^{3.8})$, a $Q^{0.1–0.5}$-fold improvement over the $O(Q^{3.9–4.3})$ scaling of published Gaussian-basis encodings [29] (corrected 2026-08-30; the retired exponent 3.15 gave $Q^{1.0–1.5}$). The load-bearing advantage is the exact Q-linearity across molecules, not this within-molecule margin.

- **QWC measurement groups:** $O(Q^{4.01})$, slightly steeper than the term count, so the shot budget for variational algorithms is *not* a source of advantage under the exact rule (re-measured 2026-08-30; the retired pair-diagonal fit gave $O(Q^{3.36})$, below the term exponent).

- **Pauli 1-norm:** $O(Q^{1.77})$ (re-measured 2026-08-30; retired: $O(Q^{1.69})$), the key result for fault-tolerant simulation. Because the 1-norm controls both Trotter step counts and qubitization query complexity, this sub-quadratic scaling implies that the GeoVac lattice basis produces Hamiltonians that are not just sparse in structure but *concentrated in weight.*

The advantage is structural: angular momentum selection rules in the Gaunt integrals enforce a decaying ERI density ($\approx M^{-0.49}$ under the exact rule), and the surviving integrals are dominated by a small number of large diagonal contributions. Equal-qubit comparisons against validated Gaussian cc-pVDZ and cc-pVTZ integrals confirm the advantage at the 1-norm level: 7.1× lower λ at $Q = 28$, translating directly into fewer Trotter steps. This sparsity is a *basis property*, present before any qubit tapering, Pauli grouping, or tensor factorization—and compatible with all of these downstream optimizations (Sec. VI D).

For quantum chemistry on quantum hardware, where circuit depth and measurement budget are the primary constraints, the choice of second-quantized basis can be as consequential as the choice of quantum algorithm. The GeoVac lattice Hamiltonian, with its $O(N)$ classical construction cost [21], $\sim Q^{3.8}$ Pauli encoding, and $O(Q^{1.77})$ simulation-cost scaling (re-measured 2026-08-30; retired:

$O(Q^{1.69})$), provides a starting point for both NISQ variational methods and fault-tolerant Hamiltonian simulation. For composed multi-center systems (Sec. IV), the pseudopotential approximation yields block-diagonal ERIs whose Pauli count factorizes exactly as $N_{\rm Pauli} = c(n_{\max})\,Q$, with a single within-molecule exponent of 3.17 for every system tested (corrected 2026-08-29; the retired rule gave 2.50–2.52). Against Gaussian baselines [29]: at $Q = 196$, composed H_2O requires $\sim 320\times$ fewer Pauli terms than the Gaussian power-law interpolation. The decisive comparison is the system-size axis, where the exact linearity in Q stands against the Gaussian $Q^{3.9\text{–}4.3}$—a gap that widens with system size, which is what makes this advantage structural and persistent. The composition architecture [23]—which decomposes each molecule into a fiber bundle of single-center orbital blocks—provides a concrete path to polyatomic systems: each additional atomic center adds an independent ERI block without creating cross-block terms, provided the pseudopotential approximation remains valid.

Software implementing the encoder (`geovac.qubit_encoding`), measurement grouping (`geovac.measurement_grouping`), and Trotter analysis (`geovac.trotter_bounds`), together with all benchmarks reported here, are available as part of the open-source GeoVac package.

a. Note added v2.0.24. The full N-electron quantum encoding comparison (Track AS, v2.0.23) has confirmed that the composed architecture is categorically sparser than direct grid encoding of the exact 4-electron angular Hamiltonian: at $Q = 10$, the full Jordan–Wigner encoding produces 3,288 Pauli terms with a 1-norm of 739 Ha, compared to 838 terms and 37 Ha for the composed encoding at $Q = 30$. *Vintage note (2026-08-30):* the composed term count is the exact-rule value (retired: 334), but the 1-norms on both sides of this comparison were measured under the retired rule and have not been re-derived—the full Jordan–Wigner side is not reproducible from the shipping library. The qualitative conclusion (the composed architecture is categorically sparser than direct grid encoding, $838 \ll 3{,}288$ at three times the qubit count) is unaffected. The root cause is structural: the full encoding maps finite-difference grid points in the 11-dimensional angular space (dense FD coupling), while the composed encoding maps orbital occupations with block-diagonal ERIs from Gaunt selection rules. Cross-pair V_{ee} costs more in Pauli terms than the PK/$Z_{\rm eff}$/exchange overhead it replaces. This validates the composed architecture as the quantum computing approach for multi-electron molecules.

b. Note added v2.0.44. Paper 20 presents the GeoVac Hamiltonian library as a practical tool for the quantum computing community, including FCI accuracy benchmarks across the LiH potential energy surface, matched-qubit (not matched-accuracy) resource comparisons against Gaussian baselines, and ecosystem export to Qiskit, OpenFermion, and PennyLane. The balanced coupled architecture (Paper 19) achieves 0.20% energy error at FCI, demonstrating that the composed partition Hamiltonian is accurate when solved exactly.

c. Note added v2.4.0. The molecule library now spans 37 systems (35 composed molecules plus the He and H_2 reference atoms), including the full first transition series of hydrides (ScH–ZnH; $Z = 21$–30, [Ar] frozen cores). All ten use isolated d-orbital valence blocks ($l_{\min} = l_{\max} = 2$) with ERI density of 4.0%, half that of $s{+}p$ blocks (8.9%). The resulting Pauli/Q ratio of 9.23 falls below the universal coefficient of 11.10 *under the retired pair-diagonal ERI rule* (Sec. III D)—an artifact of dropping more m-swap integrals at higher angular momentum, not a genuine saving (under the exact global-M_L rule the d-block is denser). Transition metals are reclassified from “out of scope” to “scoping demonstrated.”

d. Note added v2.5.0. A systematic investigation of nested (non-composed) hyperspherical encodings (Track DF, 6 sprints) established two results: (i) H-set angular momentum recoupling introduces additional ERI zeros beyond bare Gaunt selection rules, reducing ERI density by 18–26% relative to uncoupled bases (Sec. III E); (ii) the composed block-diagonal factorization is structurally necessary for heteronuclear molecular PES—single-center and mixed-exponent nested approaches produce correct binding energies but fail to reproduce equilibrium geometries (33.7% and 17.1% $R_{\rm eq}$ error for LiH). The $6j$ recoupling sparsity is basis-agnostic and applies to any encoding that uses angular momentum pair coupling.

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[1] S. McArdle, S. Endo, A. Aspuru-Guzik, S. C. Benjamin, and X. Yuan, “Quantum computational chemistry,” Rev. Mod. Phys. **92**, 015003 (2020).

[2] Y. Cao, J. Romero, J. P. Olson, *et al.*, “Quantum chemistry in the age of quantum computing,” Chem. Rev. **119**, 10856–10915 (2019).

[3] S. Bravyi, J. M. Gambetta, A. Mezzacapo, and K. Temme, “Tapering off qubits to simulate fermionic Hamiltonians,” arXiv:1701.08213 (2017).

[4] V. Verteletskyi, T.-C. Yen, and A. F. Izmaylov, “Measurement optimization in the variational quantum eigensolver using a minimum clique cover,” J. Chem. Phys. **152**, 124114 (2020).

[5] T.-C. Yen, V. Verteletskyi, and A. F. Izmaylov, “Measuring all compatible operators in one series of single-qubit measurements using unitary transformations,” J. Chem.

Theory Comput. **16**, 2400–2409 (2020).

[6] M. Motta, E. Ye, J. R. McClean, *et al.*, "Low rank representations for quantum simulation of electronic structure," npj Quantum Inf. **7**, 83 (2021).

[7] J. Lee, D. W. Berry, C. Gidney, *et al.*, "Even more efficient quantum computations of chemistry through tensor hypercontraction," PRX Quantum **2**, 030305 (2021).

[8] J. J. Goings, A. White, J. Lee, C. S. Tautermann, M. Degroote, C. Gidney, T. Shiozaki, R. Babbush, and N. C. Rubin, "Reliably assessing the electronic structure of cytochrome P450 on today's classical computers and tomorrow's quantum computers," Proc. Natl. Acad. Sci. **119**, e2203533119 (2022); arXiv:2202.01244.

[9] P. Jordan and E. Wigner, "Über das Paulische Äquivalenzverbot," Z. Phys. **47**, 631–651 (1928).

[10] S. B. Bravyi and A. Yu. Kitaev, "Fermionic quantum computation," Ann. Phys. (N.Y.) **298**, 210–226 (2002).

[11] J. R. McClean, N. C. Rubin, K. J. Sung, *et al.*, "OpenFermion: The electronic structure package for quantum computers," Quantum Sci. Technol. **5**, 034014 (2020).

[12] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, "Theory of Trotter error with commutator scaling," Phys. Rev. X **11**, 011020 (2021).

[13] E. Campbell, "Random compiler for fast Hamiltonian simulation," Phys. Rev. Lett. **123**, 070503 (2019).

[14] H. L. Tang, V. O. Shkolnikov, G. S. Barron, *et al.*, "qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor," PRX Quantum **2**, 020310 (2021).

[15] A. Kandala, A. Mezzacapo, K. Temme, *et al.*, "Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets," Nature **549**, 242–246 (2017).

[16] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory* (Dover, Mineola, NY, 1996).

[17] T. Helgaker, P. Jørgensen, and J. Olsen, *Molecular Electronic-Structure Theory* (Wiley, Chichester, 2000).

[18] W. J. Hehre, R. F. Stewart, and J. A. Pople, "Self-consistent molecular-orbital methods. I. Use of Gaussian expansions of Slater-type atomic orbitals," J. Chem. Phys. **51**, 2657–2664 (1969).

[19] J. A. Gaunt, "The triplets of helium," Philos. Trans. R. Soc. A **228**, 151–196 (1929).

[20] A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team, *NIST Atomic Spectra Database* (ver. 5.11), `https://physics.nist.gov/asd` (2024).

[21] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[22] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," GeoVac Paper 13 (2026).

[23] J. Loutey, "Composed Natural Geometries for Core-Valence Diatomics: LiH and BeH^+ from Discrete Graph Topology," GeoVac Paper 17 (2026).

[24] J. Loutey, "The Hopf Graph as a Lattice Gauge Structure: A Framework Observation," GeoVac Paper 25 (2026).

[25] J. Loutey, "Finite-Size Graph-RH for the GeoVac Hopf Graphs: Ihara Zeta, Bound-Crossing, and a Scope Boundary on Selberg-on-Hydrogen," GeoVac Paper 29 (2026).

[26] J. Burkat and N. Fitzpatrick, "The Quantum Paldus Transform: Efficient Circuits with Applications," arXiv:2506.09151 (2025).

[27] D. Rocca, C. L. Cortes, J. F. Gonthier, P. J. Ollitrault, and R. M. Parrish, "Reducing the runtime of fault-tolerant quantum simulations in chemistry through symmetry-compressed double factorization," J. Chem. Theory Comput. **20**, 4639–4653 (2024).

[28] A. E. Caesura, C. L. Cortes, W. Pol, S. Sim, M. Steudtner, G.-L. R. Anselmetti, M. Degroote, N. Moll, R. Santagati, M. Streif, and C. S. Tautermann, "Faster quantum chemistry simulations on a quantum computer with improved tensor factorization and active volume compilation," PRX Quantum **6**, 030337 (2025); arXiv:2501.06165.

[29] D. Trenev, P. J. Ollitrault, S. M. Harwood, T. P. Gujarati, S. Raman, A. Mezzacapo, and S. Mostame, "Refining resource estimation for the quantum computation of vibrational molecular spectra through Trotter error analysis," Quantum **9**, 1630 (2025); arXiv:2311.03719. [Cited for the electronic-structure Gaussian-basis Pauli counts in Table 5 (Appendix B; Jordan–Wigner with 2-qubit reduction), used as the Gaussian baseline in Table IX; the $Q^{3.92}/Q^{4.25}$ LiH/H_2O exponents and the $Q^{3.9}$–$Q^{4.3}$ span they bracket are our own log-log fit of those published counts.]

[30] J. Loutey, "Toward PK-Free Molecular Hamiltonians: Two-Center Sturmian Integrals and Coupled Composition," GeoVac Paper 19 (2026).

[31] K. G. Dyall and K. Fægri, *Introduction to Relativistic Quantum Chemistry* (Oxford University Press, Oxford, 2007).

[32] M. Reiher, N. Wiebe, K. M. Svore, D. Wecker, and M. Troyer, "Elucidating reaction mechanisms on quantum computers," Proc. Natl. Acad. Sci. USA **114**, 7555 (2017); arXiv:1605.03590.

[33] R. Szmytkowski, "Recurrence and differential relations for spherical spinors," J. Math. Chem. **42**, 397 (2007).

[34] H. A. Bethe and E. E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms* (Springer, Berlin, 1957); the Breit–Pauli Hamiltonian and the spin-orbit, spin-spin, and spin-other-orbit operators are treated canonically in Chs. 38–39.

[35] D. W. Berry, A. M. Childs, and R. Kothari, "Hamiltonian simulation with nearly optimal dependence on all parameters," in Proc. 56th IEEE FOCS (2015), pp. 792–809; arXiv:1501.01715.

[36] R. P. Martínez-y-Romero, H. N. Núñez-Yépez, and A. L. Salas-Brito, "Relativistic recursion relations for transition matrix elements," arXiv:physics/0402061 (2004).

[37] P. Chawla, V. S. Prasannaa, K. R. Swain, K. Sugisaki, and B. P. Das, "Relativistic VQE calculations of molecular electric dipole moments on trapped ion quantum hardware," Phys. Rev. A **111**, 022817 (2025); arXiv:2406.04992. [RaH 18-qubit active space: 12,556 relativistic / 2,740 non-relativistic Pauli terms (47,099 / 4,249 two-electron integrals).]

[38] E. M. Christlmaier, T. Schraivogel, P. López Ríos, A. Alavi, and D. Kats, "xTC: An efficient treatment of three-body interactions in transcorrelated methods," J. Chem. Phys. **159**, 014113 (2023).

[39] G. H. Low and Y. Su, "Quantum eigenvalue processing," SIAM J. Comput. **55**, 135 (2026); arXiv:2401.06240

(2024).
[40] A. Uvarov and A. F. Izmaylov, “Accuracy and resource advantages of quantum eigenvalue estimation with non-Hermitian transcorrelated electronic Hamiltonians,” arXiv:2511.21867 (2025).